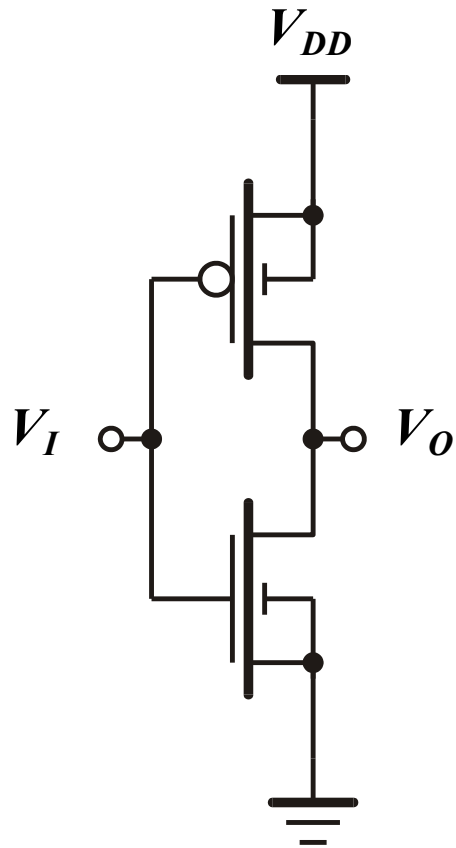


INVERTITORE CMOS STATICO

Invertitore Statico CMOS

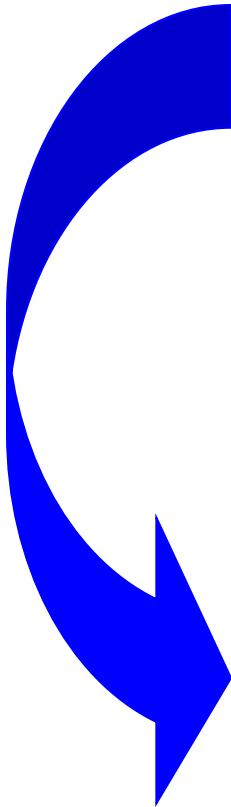
Corso di Microelettronica

Laurea Magistrale in Ingegneria Elettronica

Dipartimento di Ingegneria dell'Informazione - Università degli Studi di Brescia

Zsolt M. Kovács - Vajna

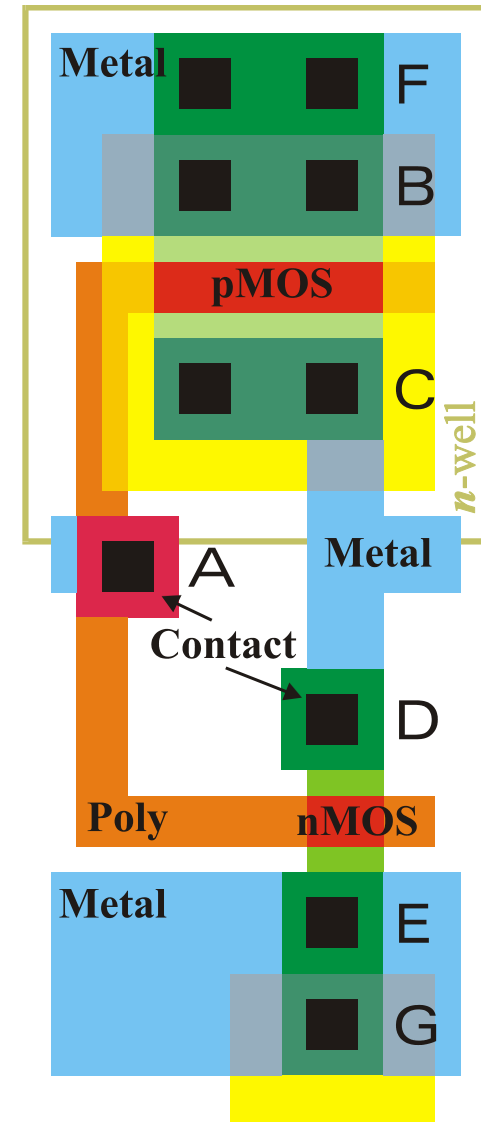
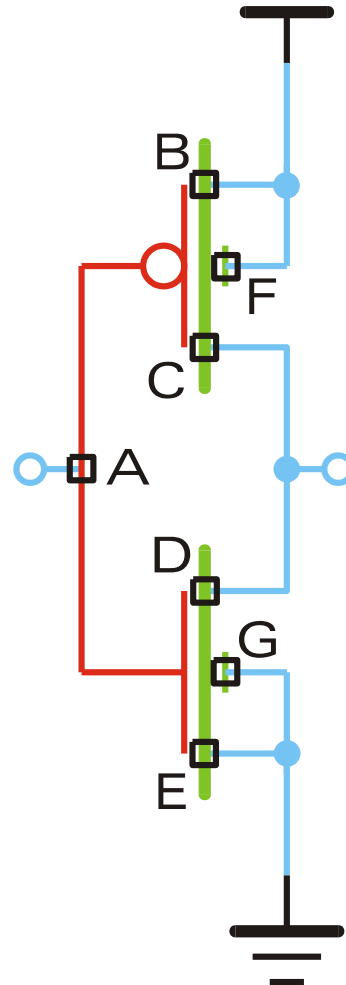
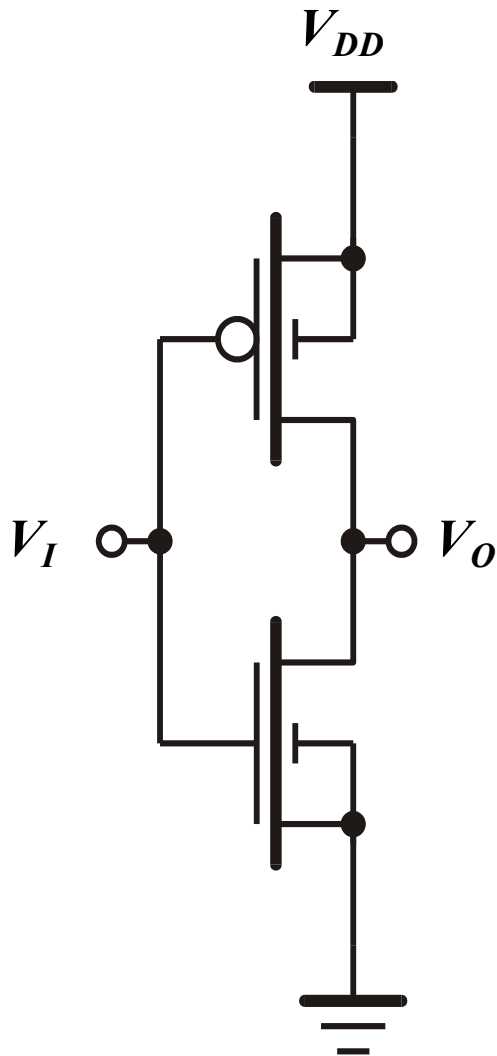
Questo materiale è riservato esclusivamente agli iscritti al Corso: la sua vendita o distribuzione, anche solo parziale e in qualunque forma e anche solo per uso interno o didattico, senza autorizzazione scritta da parte dell'autore, è vietata e sarà perseguita a norma di legge.



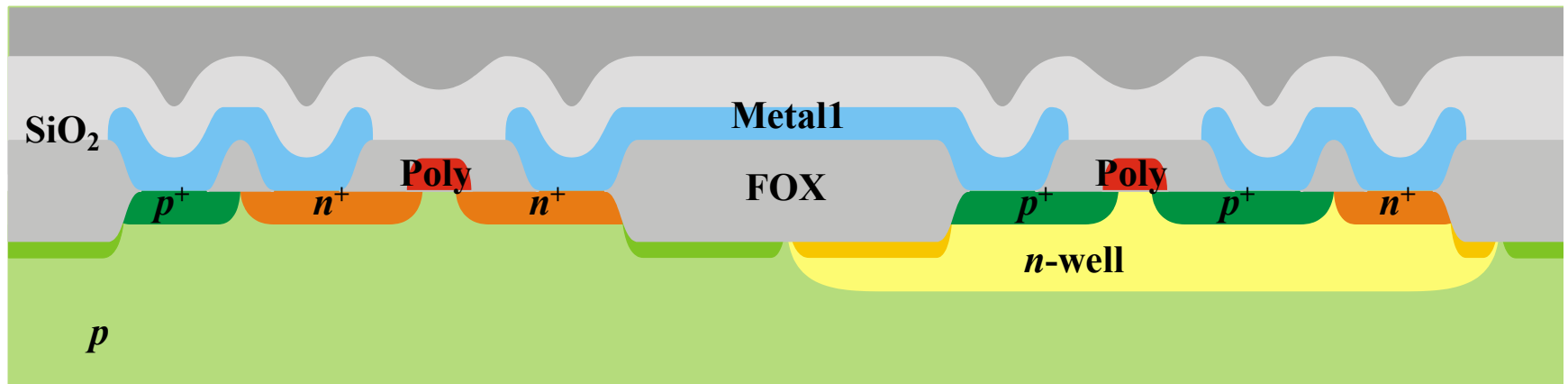
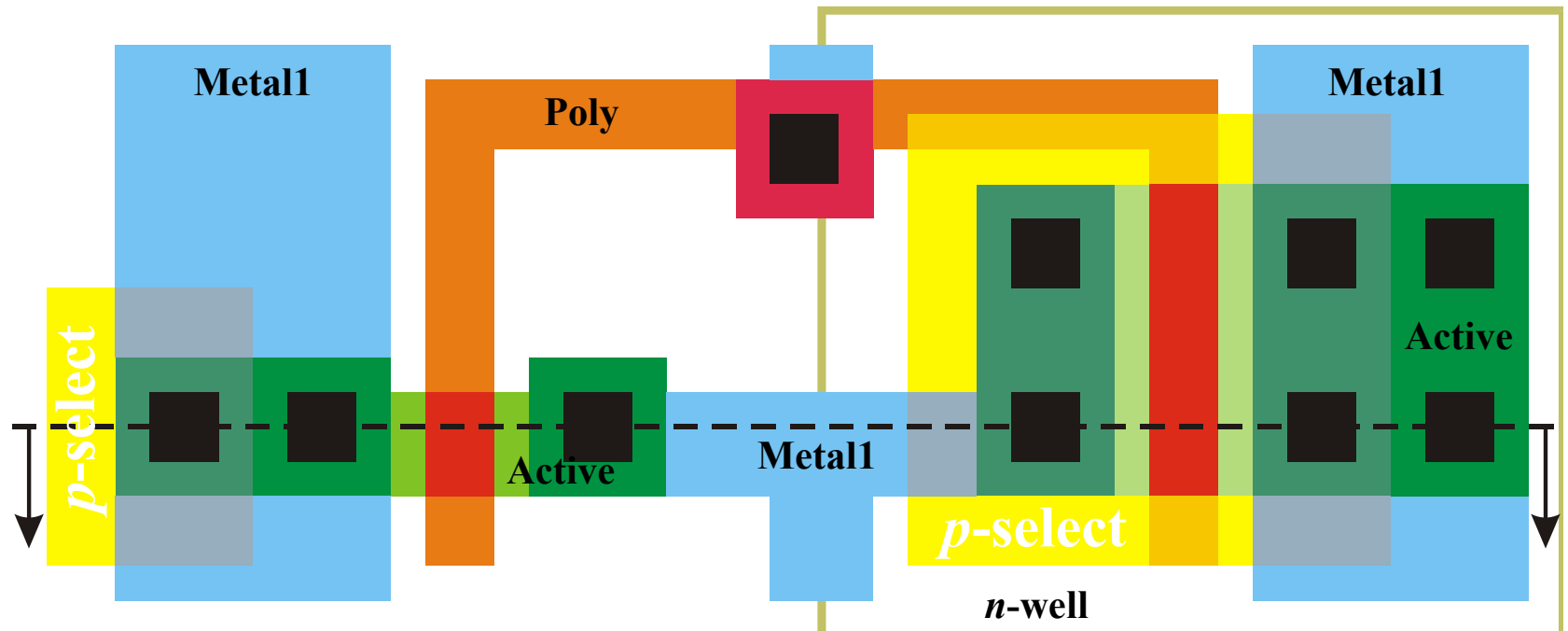
LAYOUT

- ✓ **Area occupata**
- ✓ **Margine di immunità ai disturbi**
- ✓ ***Swing* logico**
- ✓ **Soglia logica**
- ✓ **Potenza dissipata**
- ✓ **Tempi caratteristici**
- ✓ **Prodotto ritardo-consumo**

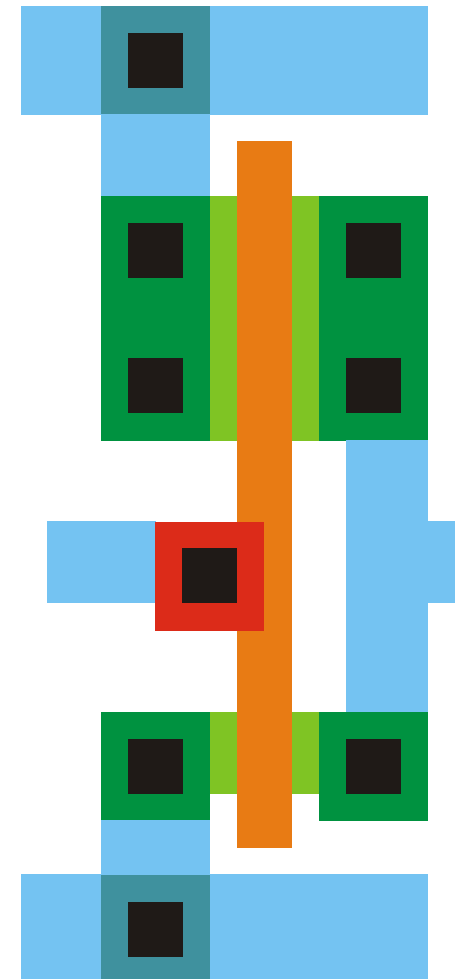
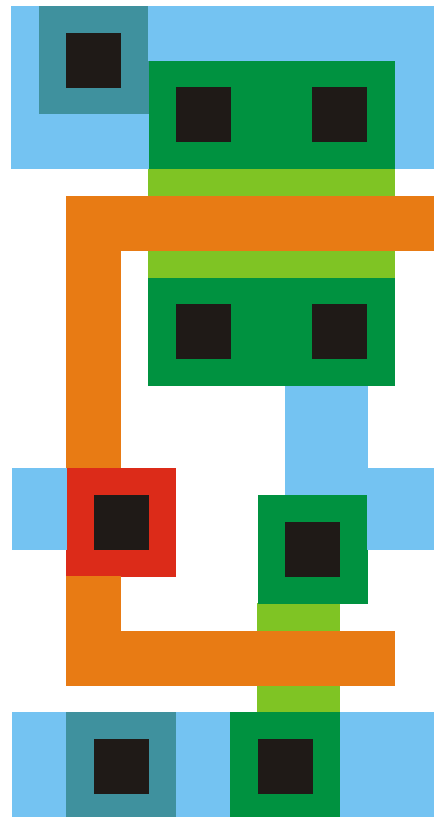
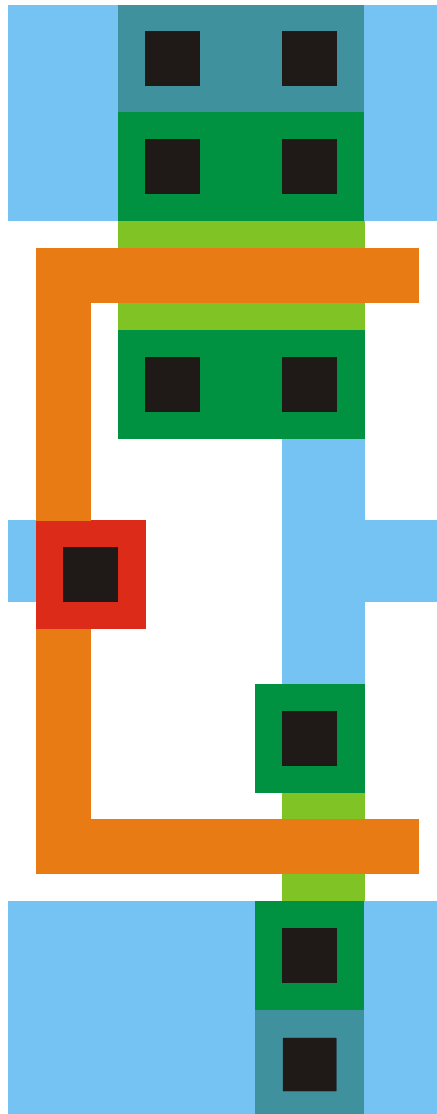
LAYOUT DELL'INVERTITORE CMOS



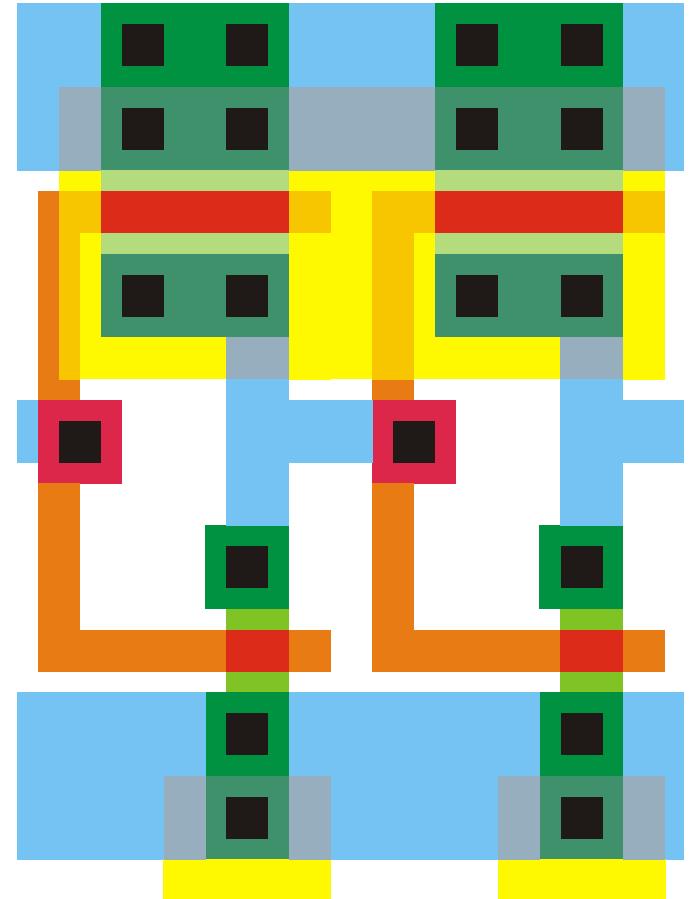
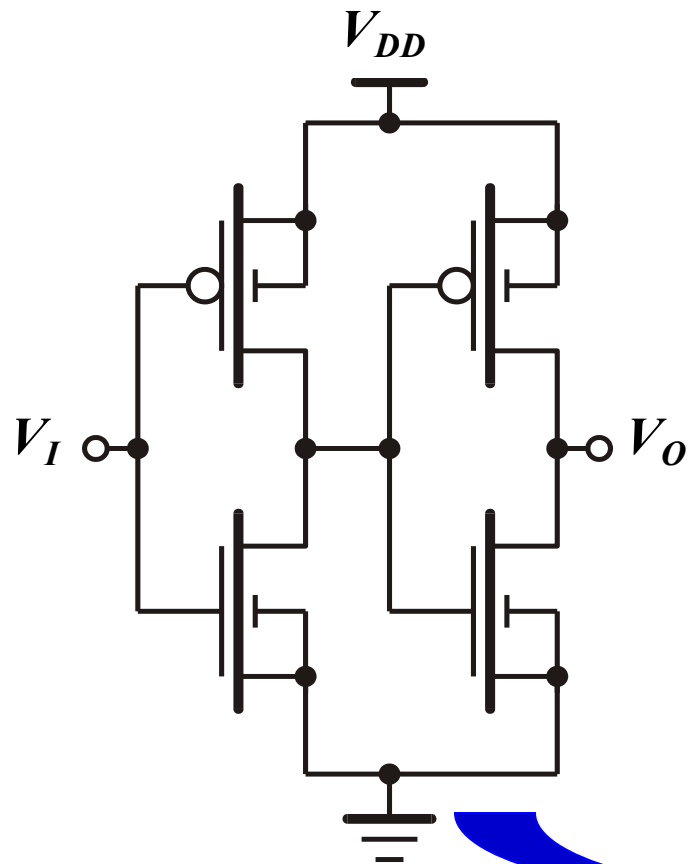
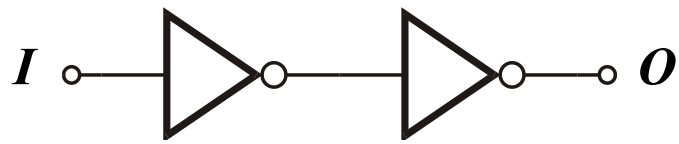
SEZIONE DELL'INVERTITORE CMOS



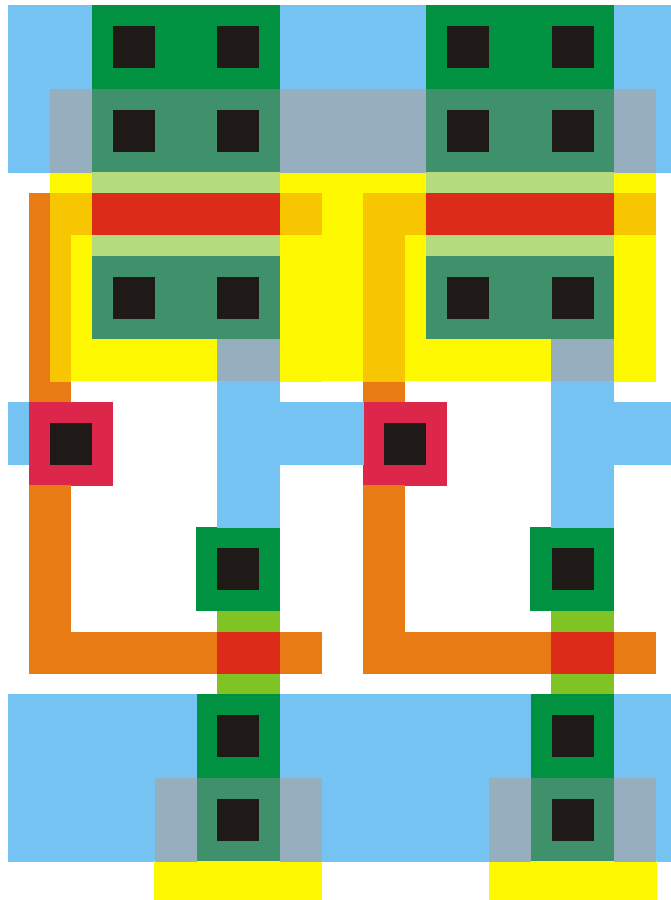
TOPOLOGIE DI INVERTITORI CMOS



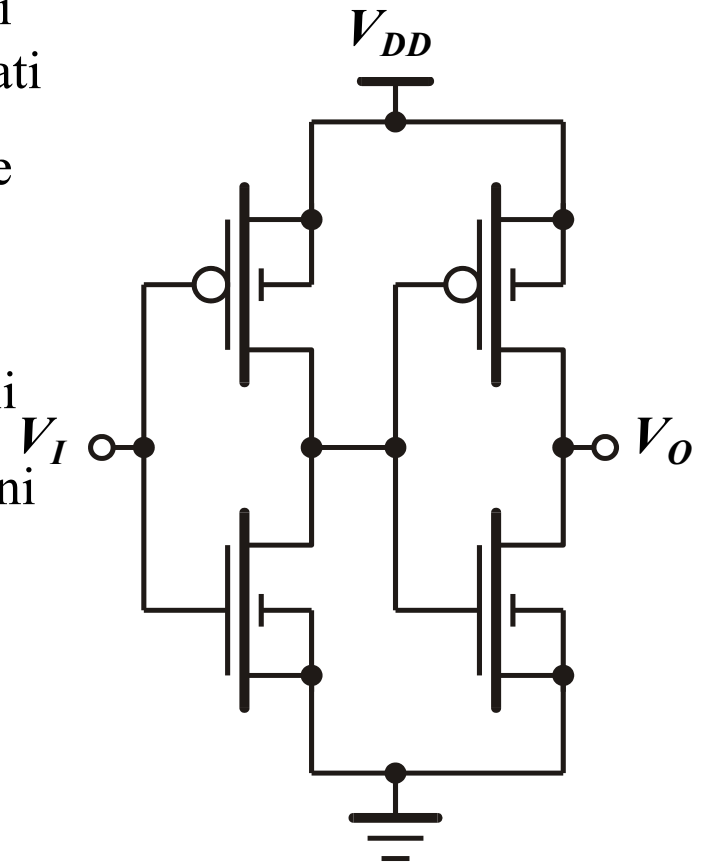
LAYOUT DI 2 INVERTORI



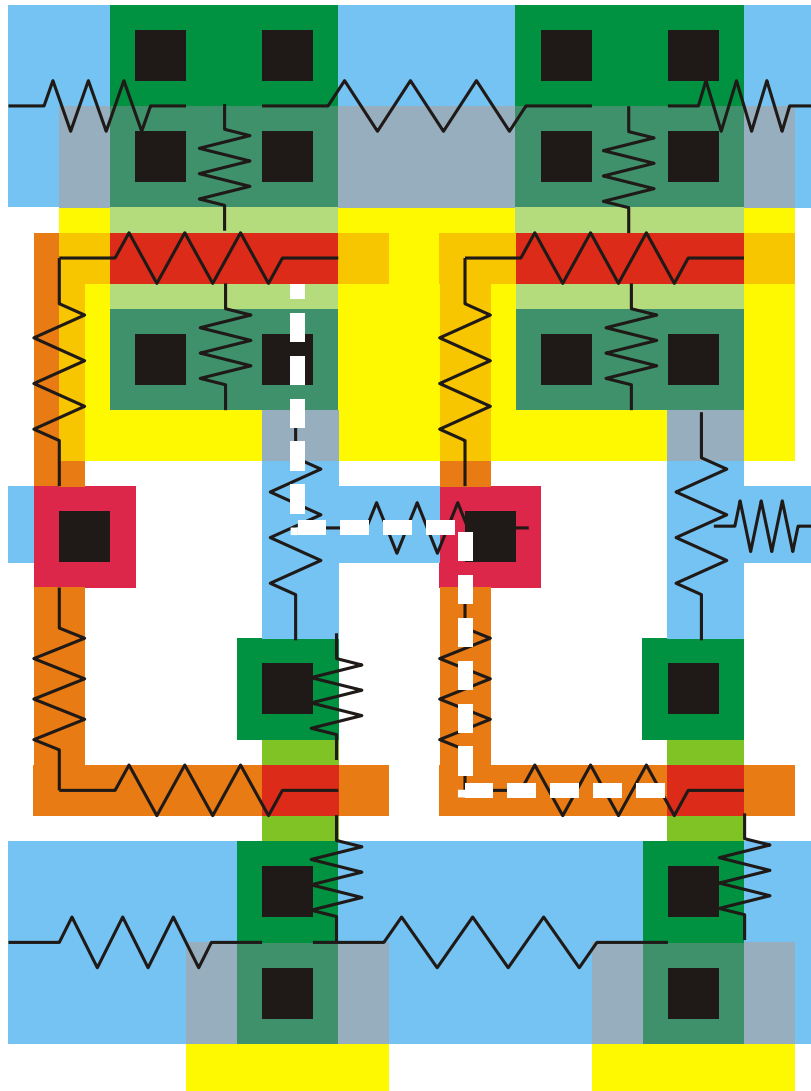
LVS (*LAYOUT* vs *SCHEMATICS*)



- Parametri concentrati
- Maschere logiche
- Materiali non ideali
- Interazioni
- ...

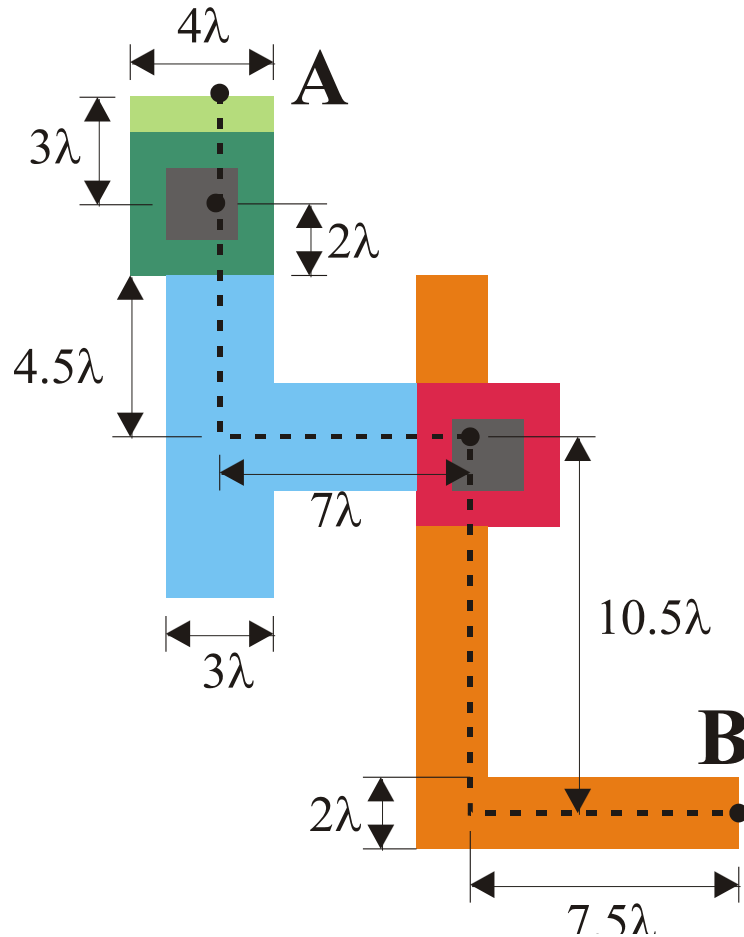


ESTRAZIONE DELLE RESISTENZE



- Parametri concentrati
- Materiali uniformi
- Densità di corrente uniformi e vettori paralleli
- SiO_2 isolante perfetto
- ...

ESEMPIO DI STIMA DI RESISTENZA



$$R_{AB} \approx R_{Dp+} \times \frac{3}{4} + 57.2\Omega$$

$$R_{cM-p+} + 36.2\Omega$$

$$R_M \times \frac{2 + 4.5 + 7}{3} + 0.2\Omega$$

$$R_{cM-P} + 25.7\Omega$$

$$R_P \times \frac{10.5 + 7.5}{2} 234.9\Omega$$

$$= 354.2\Omega$$

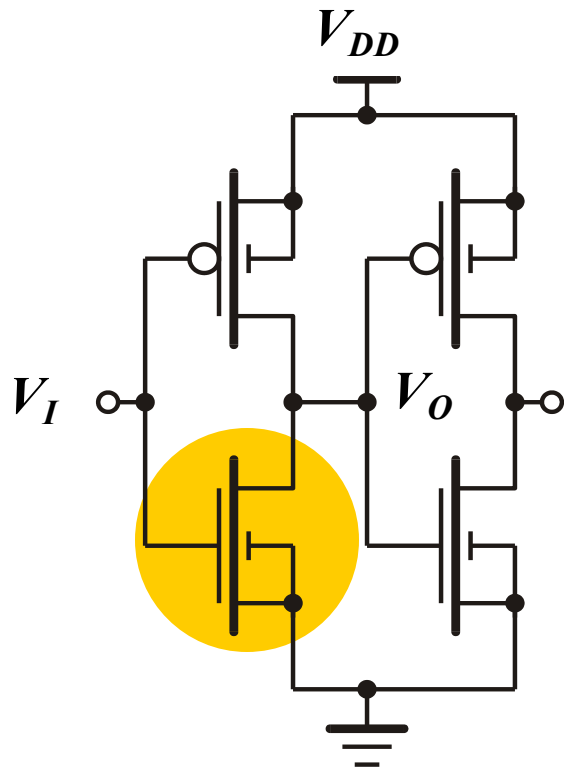
correzione degli angoli trascurata

$$R_M = 0.05\Omega/\square \quad R_P = 26.1\Omega/\square \quad (\text{AMI } 1.2\mu\text{m } 1998)$$

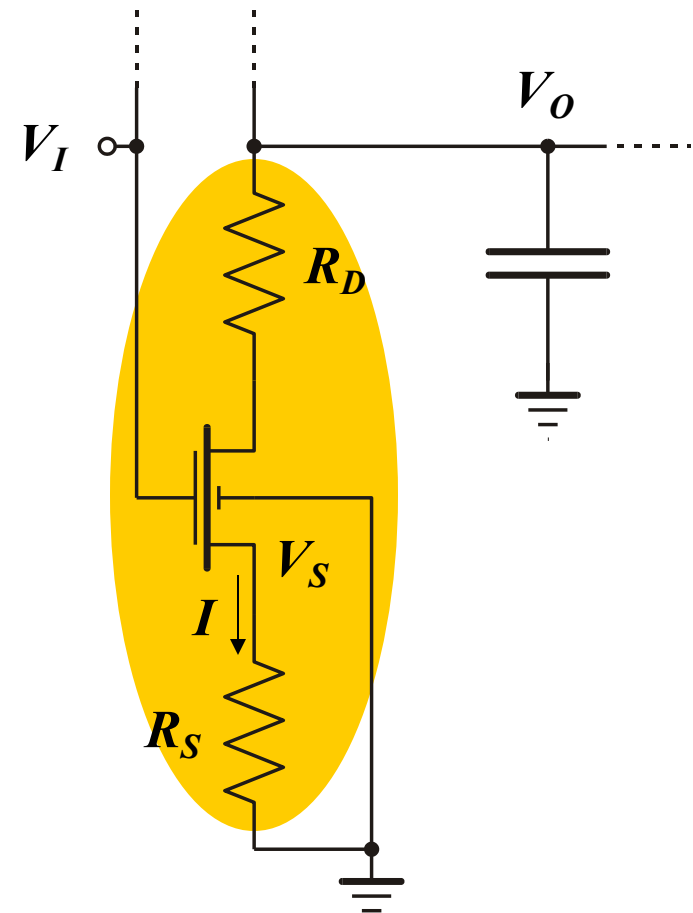
$$R_{nWell} = 1524\Omega/\square \quad R_{Dp+} = 76.2\Omega/\square \quad R_{Dn+} = 53.7\Omega/\square$$

$$R_{cM-P} = 25.7\Omega \quad R_{cM-n+} = 65.8\Omega \quad R_{cM-p+} = 36.3\Omega$$

Resistenza di Canale
(10kΩ ÷ 20kΩ)

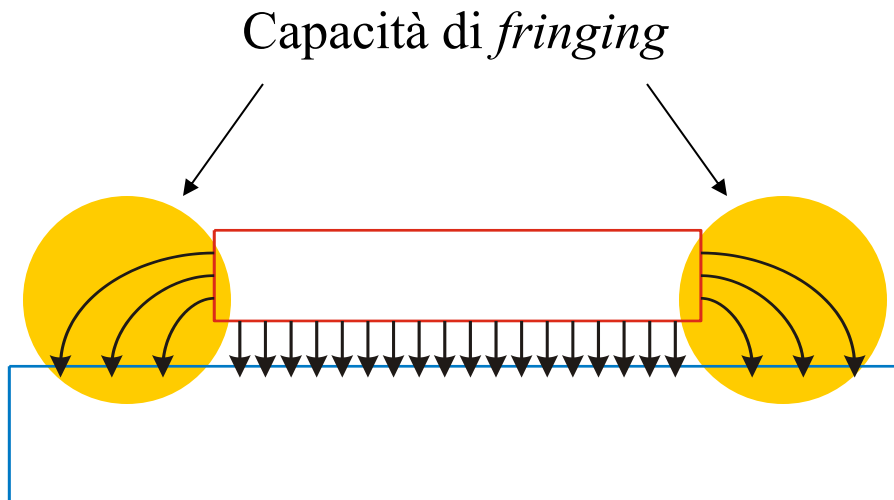


- V_{GS} cala di $R_S I$
- V_T cresce per effetto *Body*

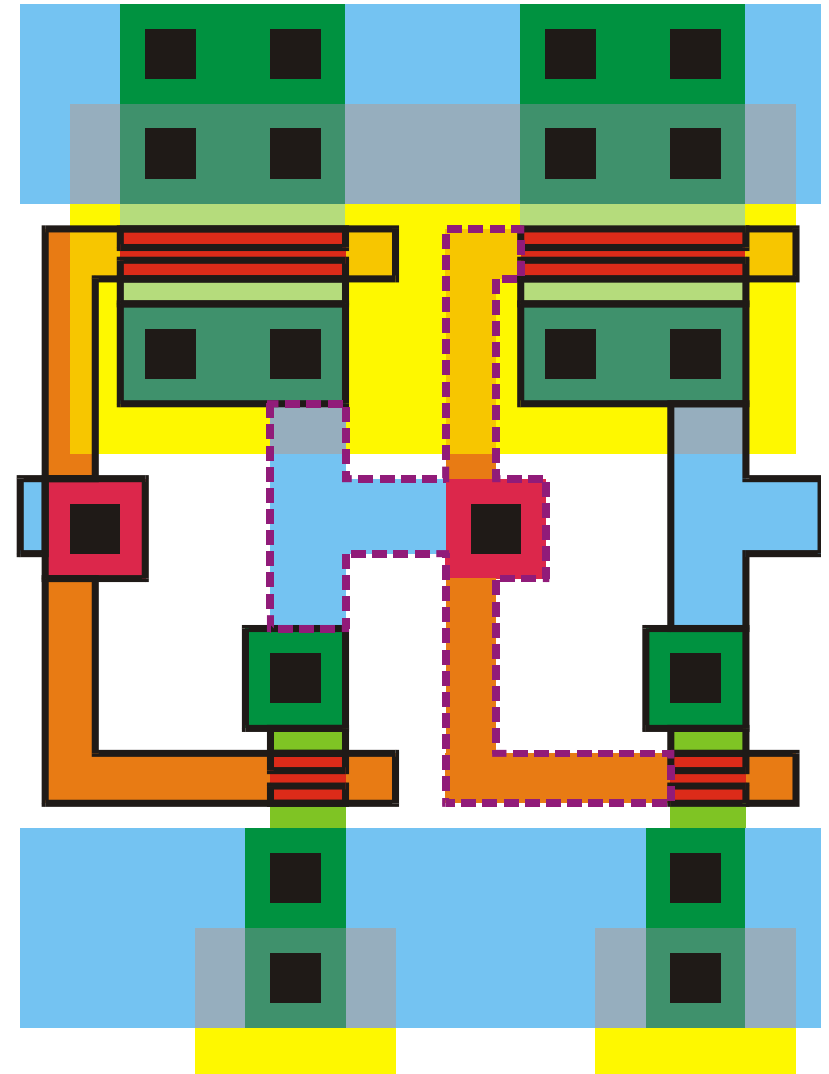


$$I_{MAX} = \frac{k_n}{2} [V_{DD} - V_S - V_{Tn}(V_S)]^2$$

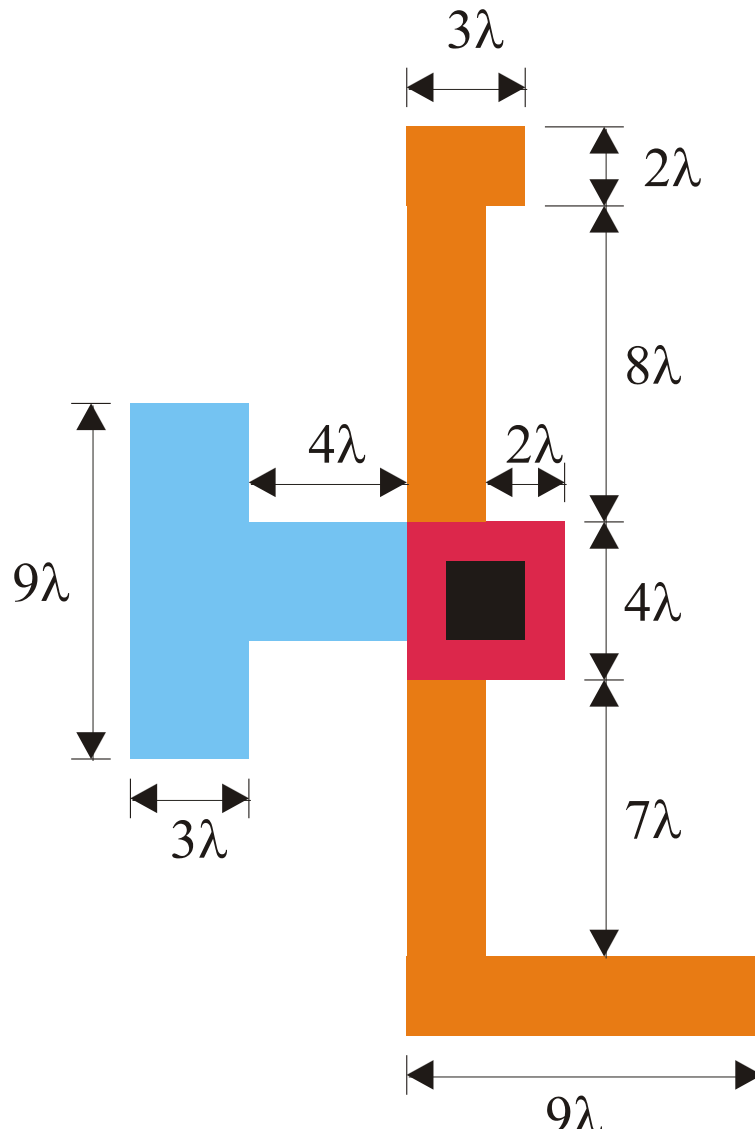
ESTRAZIONE DELLE CAPACITÀ



- ✓ Parametri concentrati
- ✓ Materiali uniformi
- ✓ Solo capacità verso substrato (e non schermato)
- ✓ Capacità di *fringing* trascurabile?
- ✓ ...



ESEMPIO DI STIMA DI CAPACITÀ



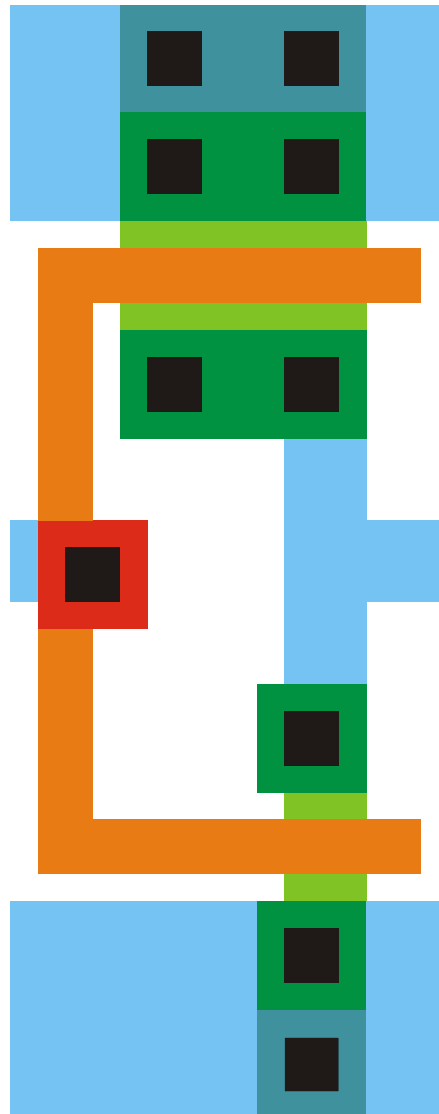
$$\begin{aligned}
 C &\approx C_M \times (9 + 4) \times 3\lambda^2 + \\
 &C_P \times (3 + 19 + 9 + 4) \times 2 \times \lambda^2 = \\
 &(C_M \times 39 + C_P \times 70) \times \lambda^2 = \\
 &(780 + 2520) \times 0.6^2 \text{ aF} = \\
 &1.19 \text{ fF}
 \end{aligned}$$

$$\begin{aligned}
 C_{Mfringing} &\approx C_{Mf} \times (9 + 6 + 8) \times \lambda = \\
 &0.52 \text{ fF}
 \end{aligned}$$

(AMI 1.2 μm 1998)

$$\begin{aligned}
 C_M &= 20 \text{ aF}/\mu\text{m}^2 & C_P &= 36 \text{ aF}/\mu\text{m}^2 \\
 C_{Dn+} &= 287 \text{ aF}/\mu\text{m}^2 & C_{Dp+} &= 296 \text{ aF}/\mu\text{m}^2 \\
 C_{Mf} &= 44 \text{ aF}/\mu\text{m}
 \end{aligned}$$

INVERTITORE MIGLIORATO

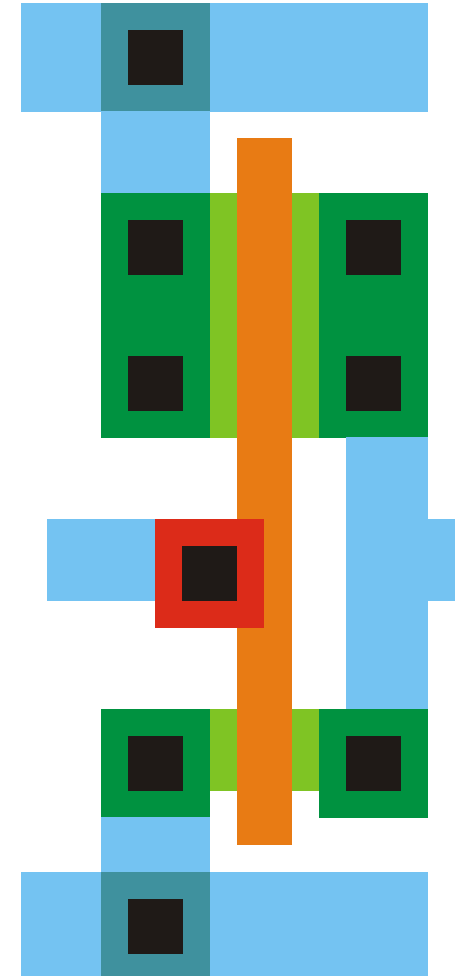


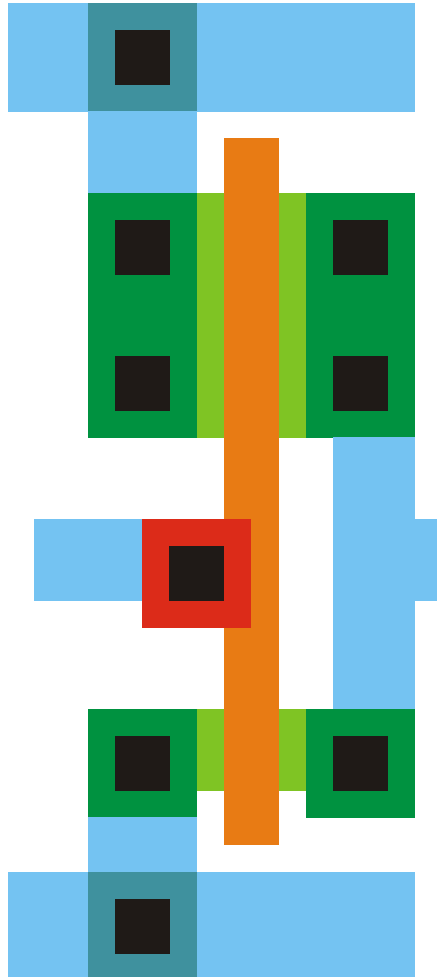
Pista in polisilicio
più corta

$$\begin{array}{ll} C_{WI} = 0.93\text{fF} & C_{WI} = 0.50\text{fF} \\ C_{WO} = 0.26\text{fF} & C_{WO} = 0.24\text{fF} \end{array}$$

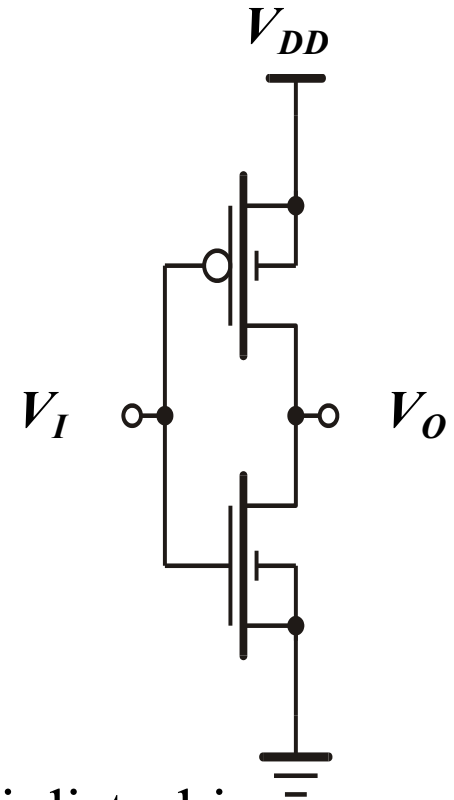


Area occupata ridotta





resistenze trascurate
< 5% del canale
dei MOS



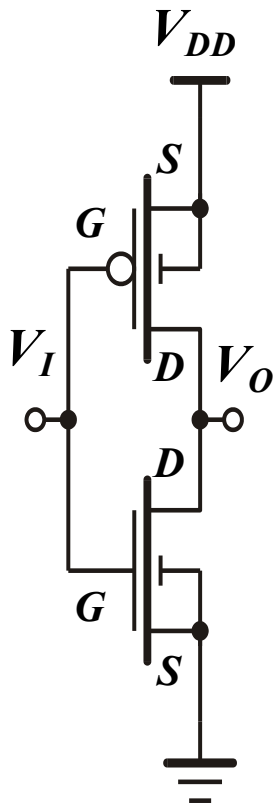
- ✓ Margine di immunità ai disturbi
- ✓ *Swing* logico
- ✓ Soglia logica
- ✓ Potenza dissipata

REGIONI DI FUNZIONAMENTO

$$V_{Tn0} = V_{Tn} = V_{Tp} = V_{Tp0}$$

$$L_n = L_p = \min$$

$$k_n = k_p$$



M_n off: ($V_{GS} < V_{Tn}$) $V_I < V_{Tn}$

lin: ($V_{GS} > V_{Tn}$) $V_I > V_{Tn}$

($V_{DS} < V_{GS} - V_{Tn}$) $V_O < V_I - V_{Tn}$

sat: ($V_{GS} > V_{Tn}$) $V_I > V_{Tn}$

($V_{DS} > V_{GS} - V_{Tn}$) $V_O > V_I - V_{Tn}$

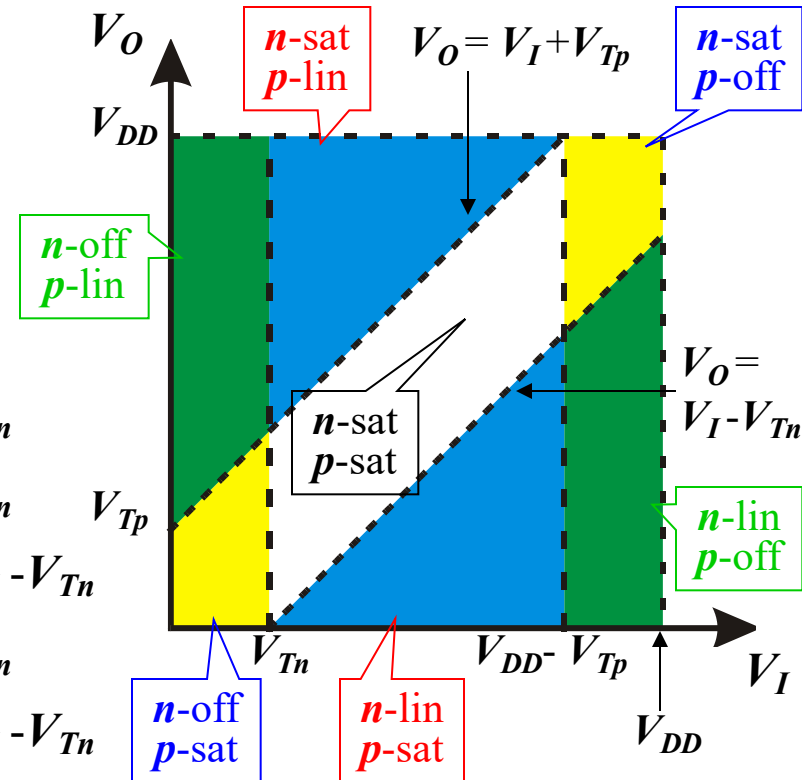
M_p off: ($V_{SG} < V_{Tp}$) $V_{DD} - V_I < V_{Tp}$ $V_I > V_{DD} - V_{Tp}$

lin: ($V_{SG} > V_{Tp}$) $V_{DD} - V_I > V_{Tp}$ $V_I < V_{DD} - V_{Tp}$

($V_{SD} < V_{SG} - V_{Tp}$) $V_{DD} - V_O < V_{DD} - V_I - V_{Tp}$ $V_O > V_I + V_{Tp}$

sat: ($V_{SG} > V_{Tp}$) $V_{DD} - V_I > V_{Tp}$ $V_I < V_{DD} - V_{Tp}$

($V_{SD} > V_{SG} - V_{Tp}$) $V_{DD} - V_O > V_{DD} - V_I - V_{Tp}$ $V_O < V_I + V_{Tp}$



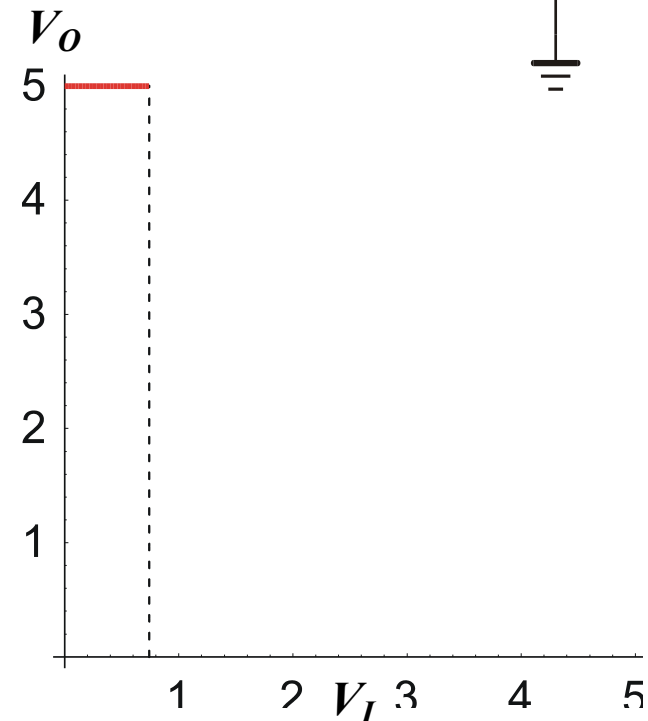
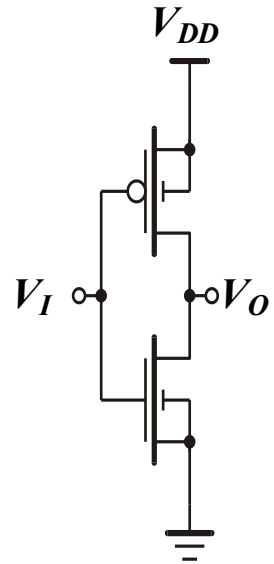
$$V_I \in [0, V_{Tn}]$$

$$I_{DSn} = I_{SDp}$$

$$\begin{cases} I_{DSn} = 0 \\ I_{SDp} = k_p \left[(V_{SGp} - V_{Tp}) V_{SDp} - \frac{V_{SDp}^2}{2} \right] \end{cases}$$

$$0 = k_p \left[(V_{DD} - V_I - V_{Tp})(V_{DD} - V_O) - \frac{(V_{DD} - V_O)^2}{2} \right]$$

$$V_O = V_{DD}$$



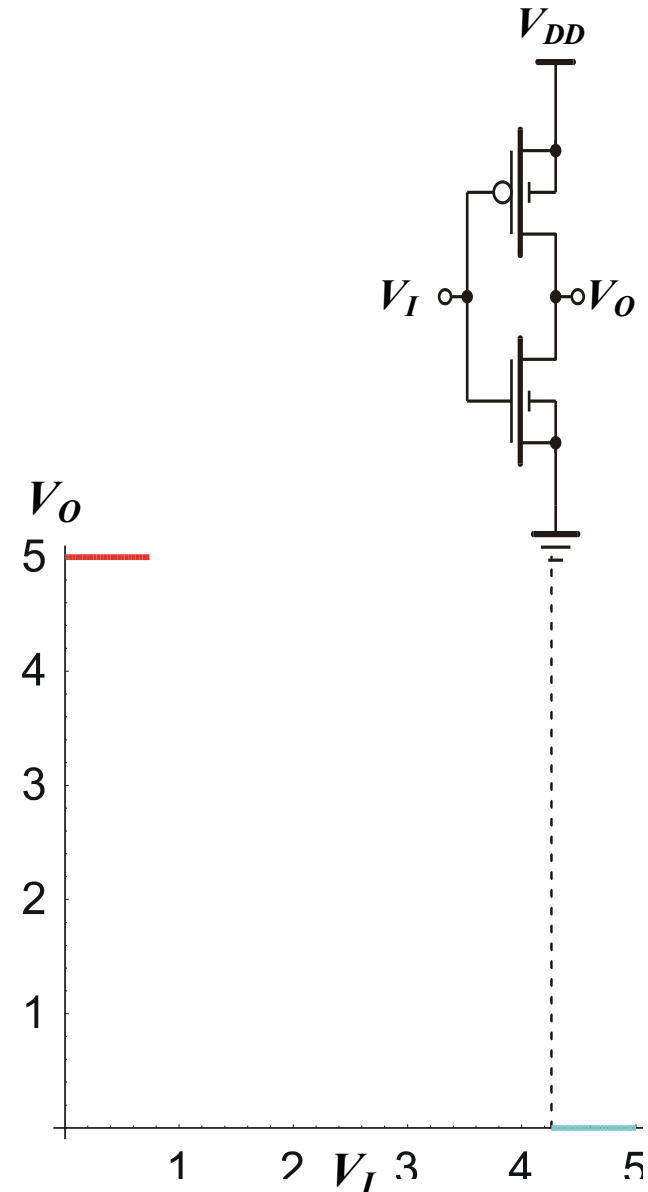
$$V_I \in [V_{DD} - V_{Tp}, V_{DD}]$$

$$I_{DSn} = I_{SDp}$$

$$\begin{cases} I_{DSn} = k_n \left[(V_{GSn} - V_{Tn}) V_{DSn} - \frac{V_{DSn}^2}{2} \right] \\ I_{SDp} = 0 \end{cases}$$

$$k_n \left[(V_I - V_{Tn}) V_o - \frac{V_o^2}{2} \right] = 0$$

$$V_o = 0$$



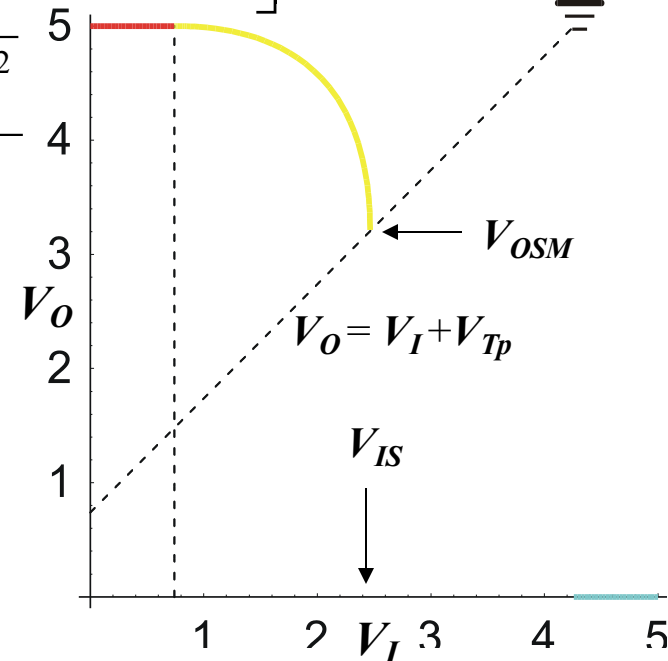
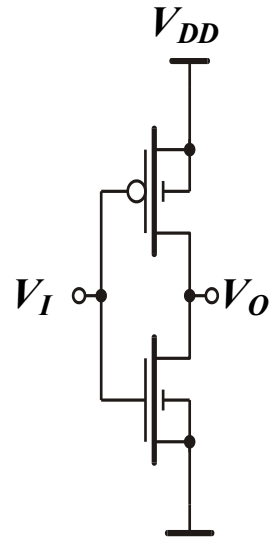
$$V_I \in [V_{Tn}, V_{IS}] \quad \left\{ \begin{array}{l} I_{DSn} = \frac{k_n}{2} (V_{GSn} - V_{Tn})^2 \\ I_{SDp} = k_p \left[(V_{SGp} - V_{Tp}) V_{SDp} - \frac{V_{SDp}^2}{2} \right] \end{array} \right.$$

$$\frac{k_n}{2} (V_I - V_{Tn})^2 = k_p \left[(V_{DD} - V_I - V_{Tp})(V_{DD} - V_O) - \frac{(V_{DD} - V_O)^2}{2} \right]$$

$$V_O = V_I + V_{Tp} + \frac{\sqrt{k_p} (V_{DD} - V_I - V_{Tp})^2 - k_n (V_I - V_{Tn})^2}{\sqrt{k_p}}$$

$$V_{IS} = \frac{\sqrt{k_n} V_{Tn} + \sqrt{k_p} (V_{DD} - V_{Tp})}{\sqrt{k_n} + \sqrt{k_p}} = 2.46V$$

$$V_{OSM} = \frac{\sqrt{k_n} (V_{Tn} + V_{Tp}) + \sqrt{k_p} V_{DD}}{\sqrt{k_n} + \sqrt{k_p}} = 3.20V$$



$$V_I \in [V_{IS}, V_{DD} - V_{Tp}] \quad \left\{ \begin{array}{l} I_{DSn} = k_n \left[(V_{GSn} - V_{Tn}) V_{DSn} - \frac{V_{DSn}^2}{2} \right] \\ I_{SDp} = \frac{k_p}{2} (V_{SGp} - V_{Tp})^2 \end{array} \right.$$

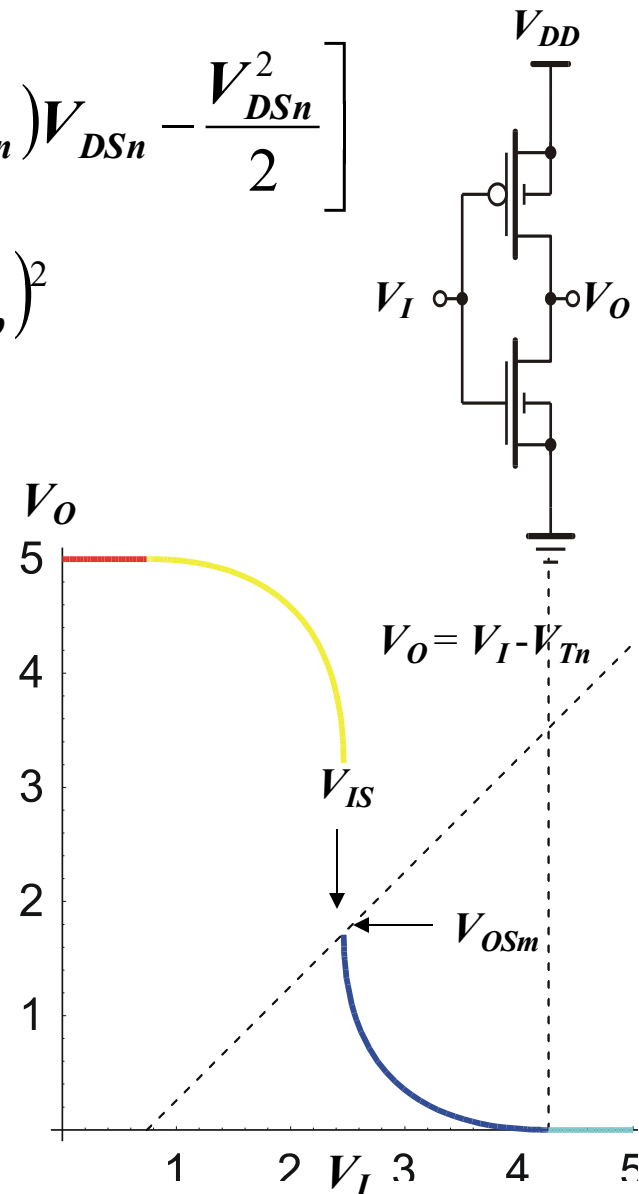
$$I_{DSn} = I_{SDp}$$

$$k_n \left[(V_I - V_{Tn}) V_O - \frac{V_O^2}{2} \right] = \frac{k_p}{2} (V_{DD} - V_I - V_{Tp})^2$$

$$V_O = V_I - V_{Tn} - \frac{\sqrt{k_n (V_I - V_{Tn})^2 - k_p (V_{DD} - V_I - V_{Tp})^2}}{\sqrt{k_n}}$$

$$V_{IS} = \frac{\sqrt{k_n} V_{Tn} + \sqrt{k_p} (V_{DD} - V_{Tp})}{\sqrt{k_n} + \sqrt{k_p}} = 2.46V$$

$$V_{OSm} = \frac{\sqrt{k_p} (V_{DD} - V_{Tn} - V_{Tp})}{\sqrt{k_n} + \sqrt{k_p}} = 1.72V$$



$$V_I = V_{IS}$$

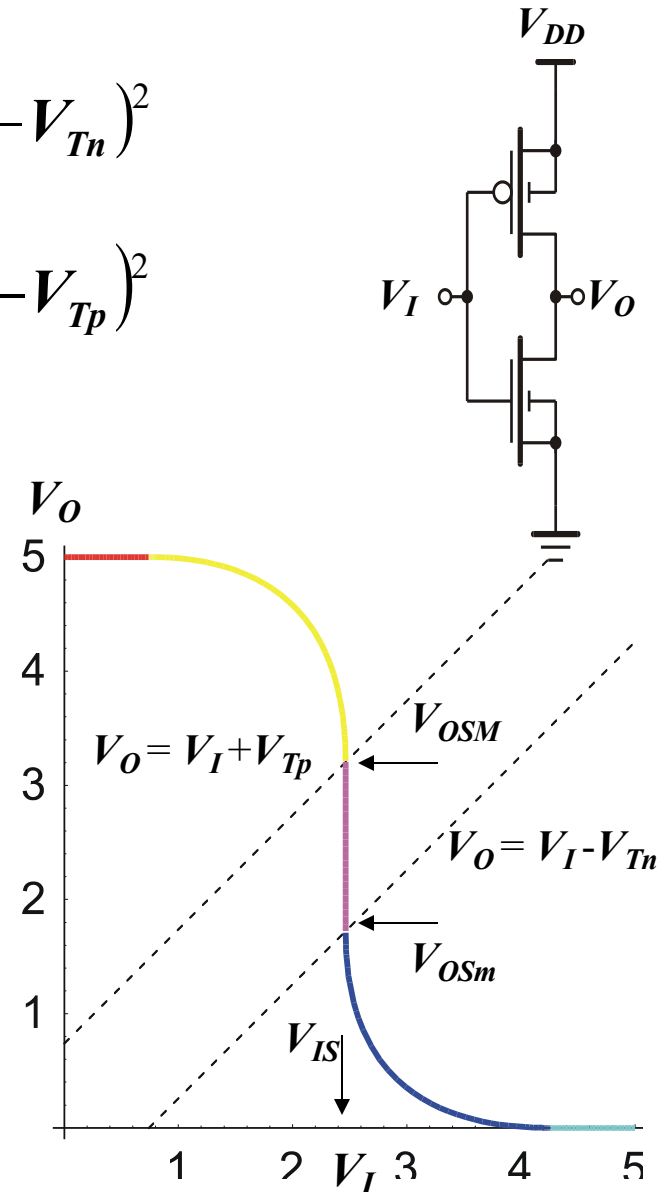
$$I_{DSn} = I_{SDp} \quad \left\{ \begin{array}{l} I_{DSn} = \frac{k_n}{2} (V_{GSn} - V_{Tn})^2 \\ I_{SDp} = \frac{k_p}{2} (V_{SGp} - V_{Tp})^2 \end{array} \right.$$

$$\frac{k_n}{2} (V_I - V_{Tn})^2 = \frac{k_p}{2} (V_{DD} - V_I - V_{Tp})^2$$

$$V_{IS} = \frac{\sqrt{k_n} V_{Tn} + \sqrt{k_p} (V_{DD} - V_{Tp})}{\sqrt{k_n} + \sqrt{k_p}} = 2.46\text{V}$$

$$V_{OSM} - V_{OSm} = V_{Tn} + V_{Tp} = 1.48\text{V}$$

$$A_v(V_{IS}) = \left. \frac{dV_o}{dV_I} \right|_{V_I=V_{IS}} \rightarrow -\infty$$



$k_n' = 80\mu\text{AV}^{-2}$	$k_p' = 27\mu\text{AV}^{-2}$
$V_{Tn0} = 740\text{mV}$	$V_{Tp0} = 740\text{mV}$
$W_n = 1.80\mu\text{m}$	$W_p = 5.33\mu\text{m}$
$L_n = 1.20\mu\text{m}$	$L_p = 1.20\mu\text{m}$
$\lambda_n = 0.037\text{V}^{-1}$	$\lambda_p = 0.061\text{V}^{-1}$

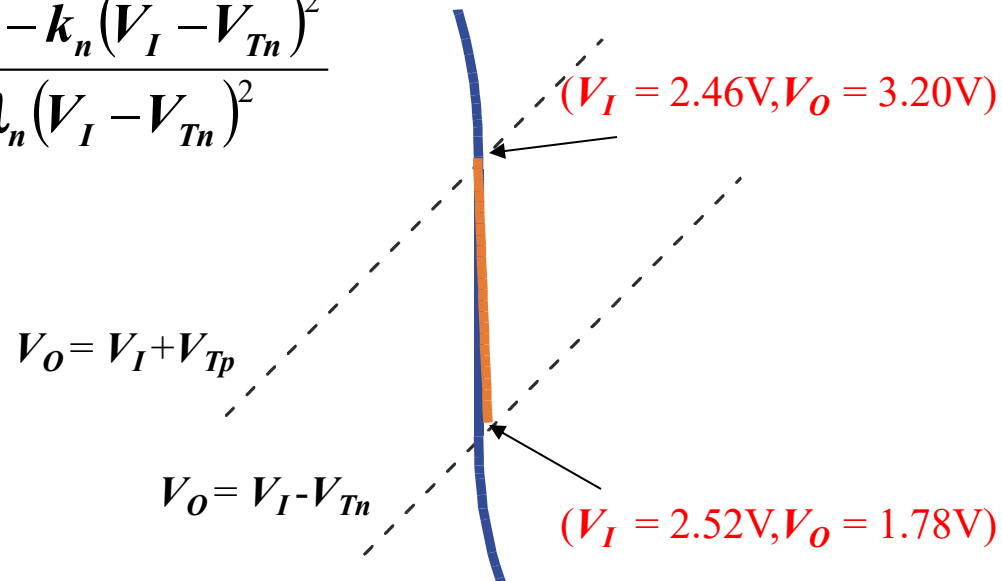
$$\begin{cases} I_{DSn} = \frac{k_n}{2} (V_{GSn} - V_{Tn})^2 (1 + \lambda_n V_{DSn}) \\ I_{SDp} = \frac{k_p}{2} (V_{SGp} - V_{Tp})^2 (1 + \lambda_p V_{SDp}) \end{cases}$$

$$\frac{k_n}{2} (V_I - V_{Tn})^2 (1 + \lambda_n V_O) = \frac{k_p}{2} (V_{DD} - V_I - V_{Tp})^2 [1 + \lambda_p (V_{DD} - V_O)]$$

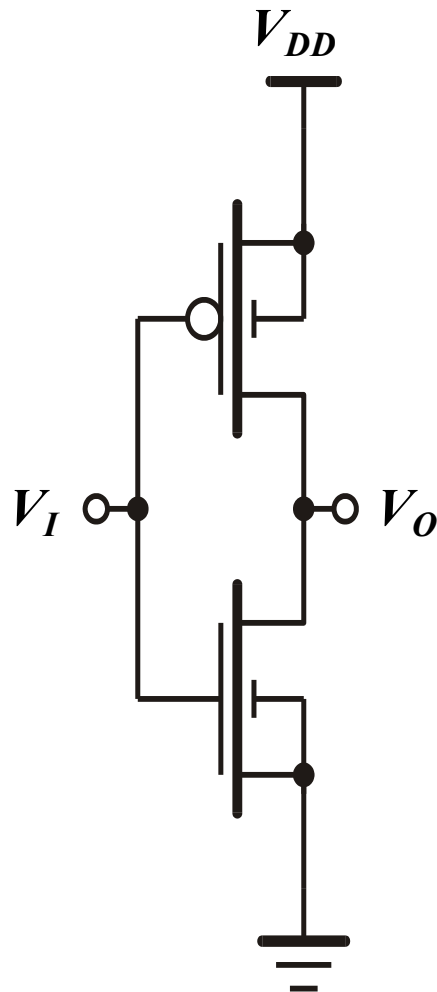
$$V_O = \frac{k_p (1 + \lambda_p V_{DD}) (V_{DD} - V_I - V_{Tp})^2 - k_n (V_I - V_{Tn})^2}{k_p \lambda_p (V_{DD} - V_I - V_{Tp})^2 - k_n \lambda_n (V_I - V_{Tn})^2}$$

$$V_{IS} = 2.46\text{V}$$

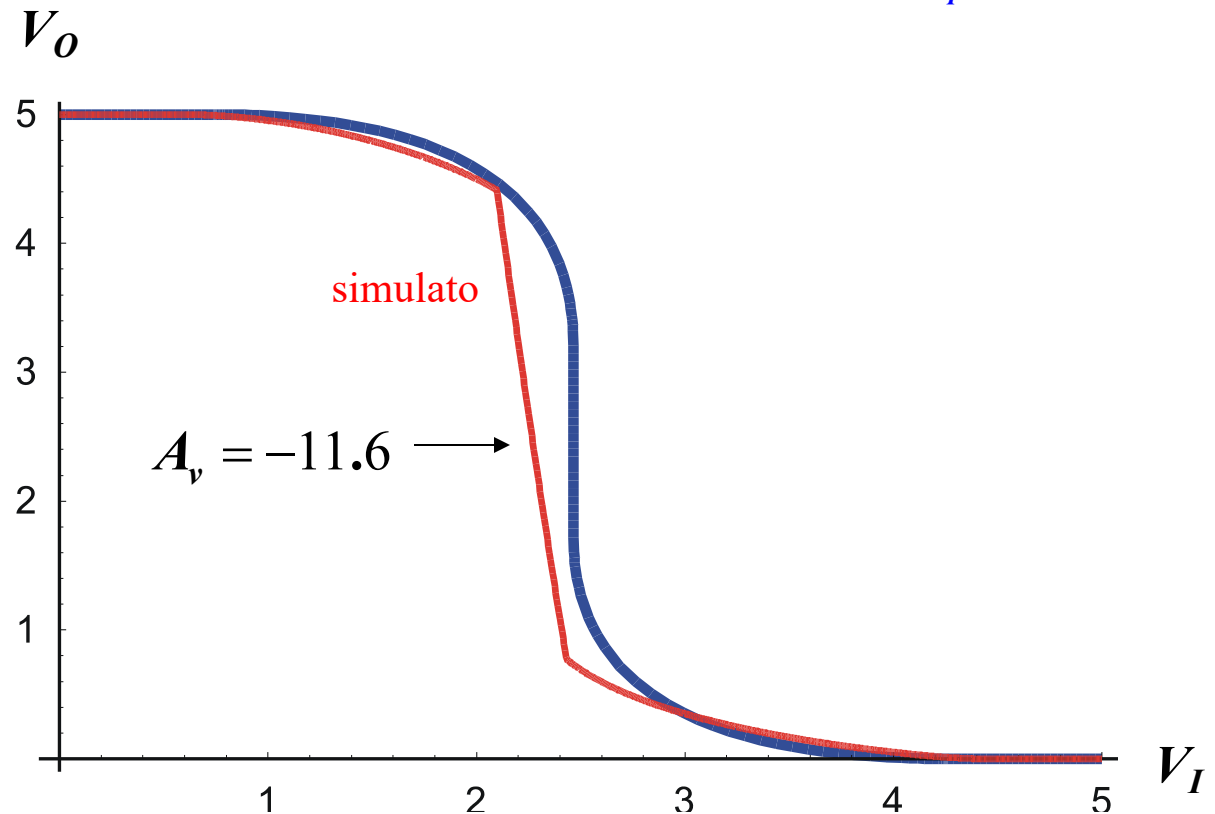
$$A_v(V_{IS}) = \left. \frac{dV_O}{dV_I} \right|_{V_I=V_{IS}} = -25.9$$

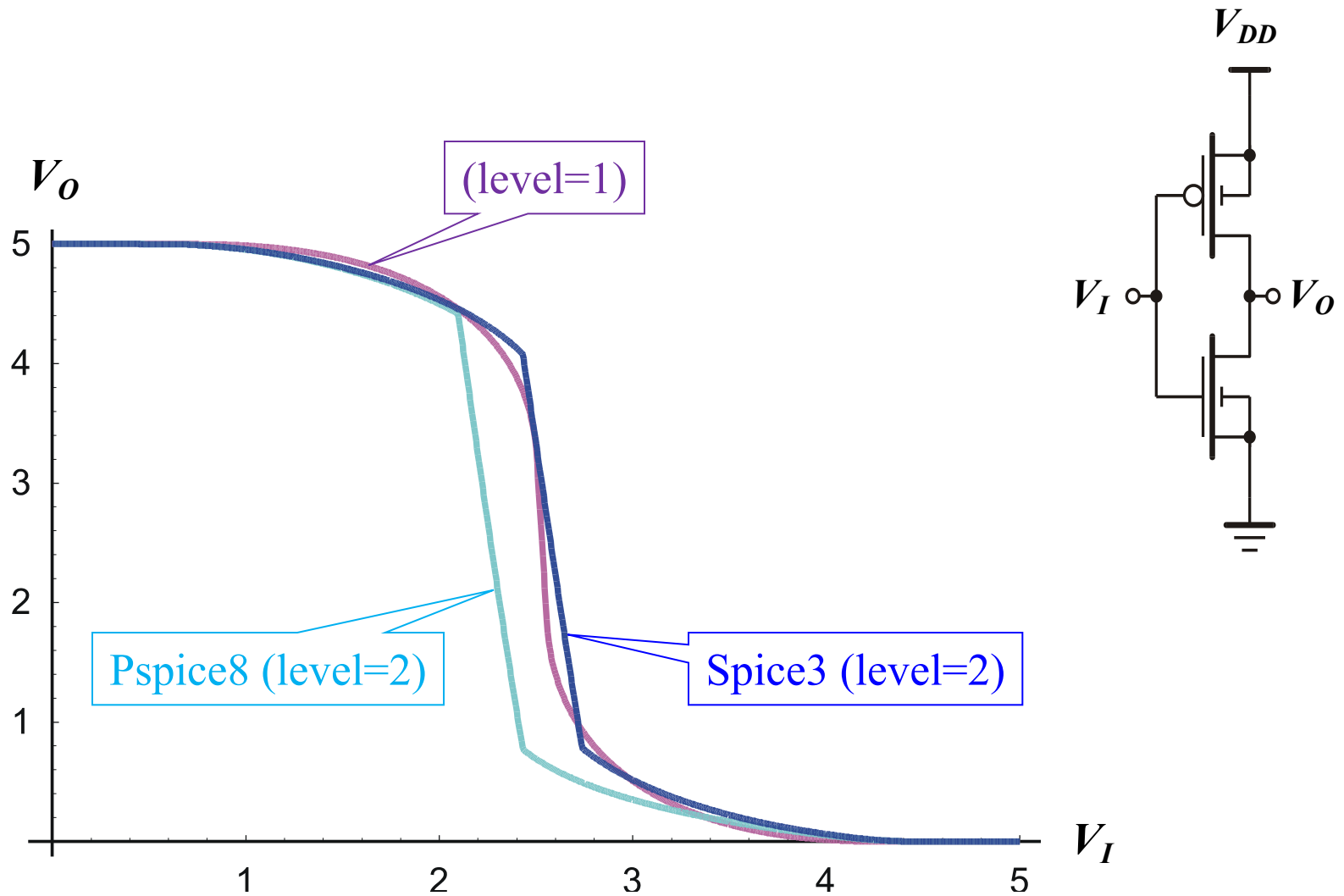


CAR. DI TRASFERIMENTO

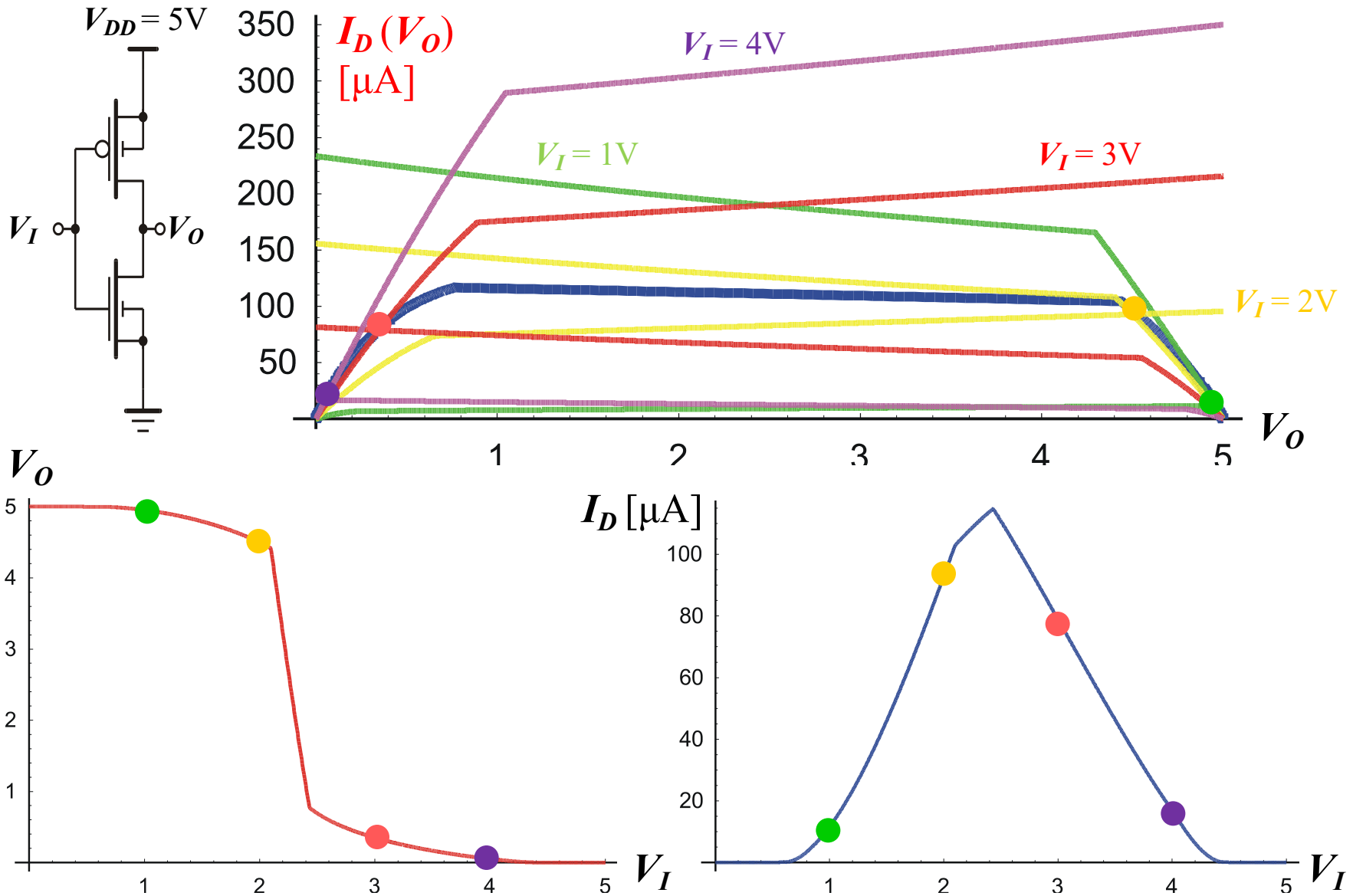


$k_n' = 80\mu\text{AV}^{-2}$	$k_p' = 27\mu\text{AV}^{-2}$
$V_{Tn0} = 740\text{mV}$	$V_{Tp0} = 740\text{mV}$
$W_n = 1.80\mu\text{m}$	$W_p = 5.33\mu\text{m}$
$L_n = 1.20\mu\text{m}$	$L_p = 1.20\mu\text{m}$
$\lambda_n = 0\text{V}^{-1}$	$\lambda_p = 0\text{V}^{-1}$



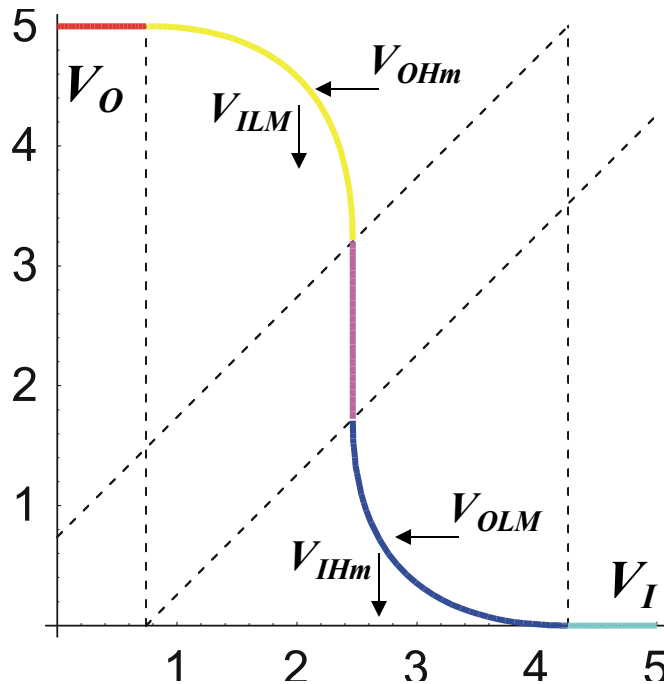


$I_D(V_I), I_D(V_O)$



CALCOLO DI NM

$$\begin{aligned} k_n' &= 80 \mu\text{A/V}^2 & k_p' &= 27 \mu\text{A/V}^2 \\ V_{Tn0} &= 740 \text{mV} & V_{Tp0} &= 740 \text{mV} \\ W_n &= 1.80 \mu\text{m} & W_p &= 5.33 \mu\text{m} \\ L_n &= 1.20 \mu\text{m} & L_p &= 1.20 \mu\text{m} \\ \lambda_n &= 0 \text{V}^{-1} & \lambda_p &= 0 \text{V}^{-1} \end{aligned}$$



$$NM_H = V_{OHm} - V_{IHm} = 1.67 \text{V}$$

$$NM_L = V_{ILM} - V_{OLM} = 1.57 \text{V}$$

$$NM = \min\{NM_L, NM_H\} = 1.57 \text{V}$$

$$V_O = V_I + V_{Tp} + \frac{\sqrt{k_p(V_{DD} - V_I - V_{Tp})^2 - k_n(V_I - V_{Tn})^2}}{\sqrt{k_p}}$$

$$\frac{\partial V_O}{\partial V_I} = 1 - \frac{k_n(V_I - V_{Tn}) + k_p(V_{DD} - V_I - V_{Tp})}{\sqrt{k_p} \sqrt{k_p(V_{DD} - V_I - V_{Tp})^2 - k_n(V_I - V_{Tn})^2}}$$

$$\frac{\partial V_O}{\partial V_I} = -1 \Rightarrow V_{ILM} = 2.01 \text{V} \quad V_{OHm} = 4.57 \text{V}$$

$$V_O = V_I - V_{Tn} - \frac{\sqrt{k_n(V_I - V_{Tn})^2 - k_p(V_{DD} - V_I - V_{Tp})^2}}{\sqrt{k_n}}$$

$$\frac{\partial V_O}{\partial V_I} = 1 - \frac{k_n(V_I - V_{Tn}) + k_p(V_{DD} - V_I - V_{Tp})}{\sqrt{k_n} \sqrt{k_n(V_I - V_{Tn})^2 - k_p(V_{DD} - V_I - V_{Tp})^2}}$$

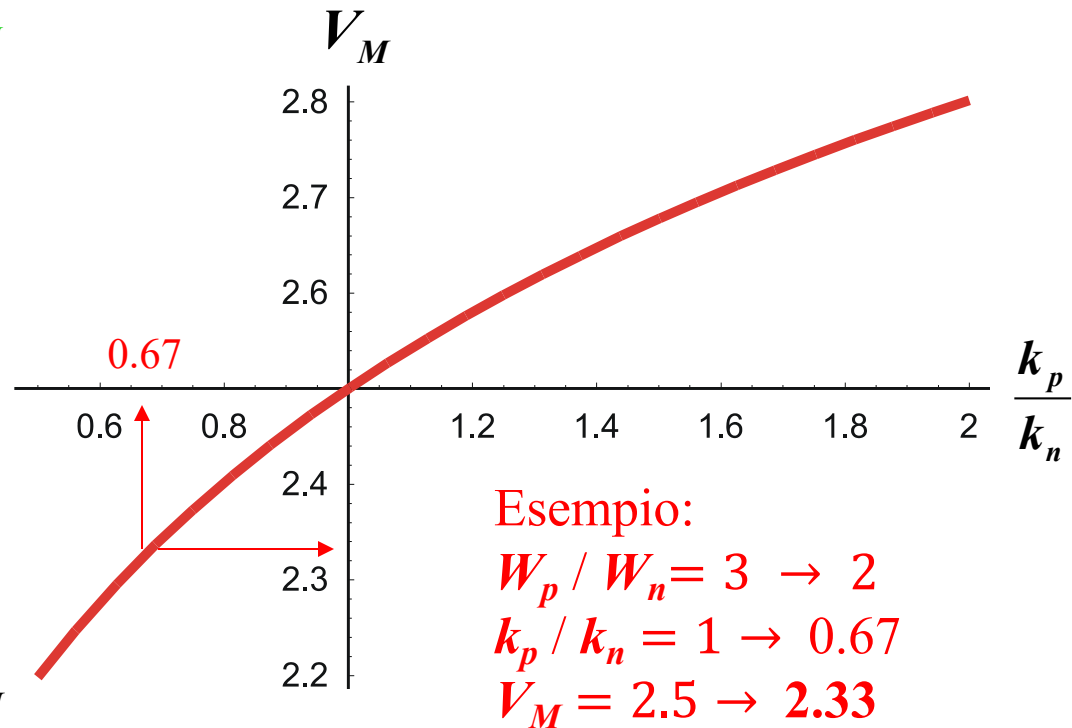
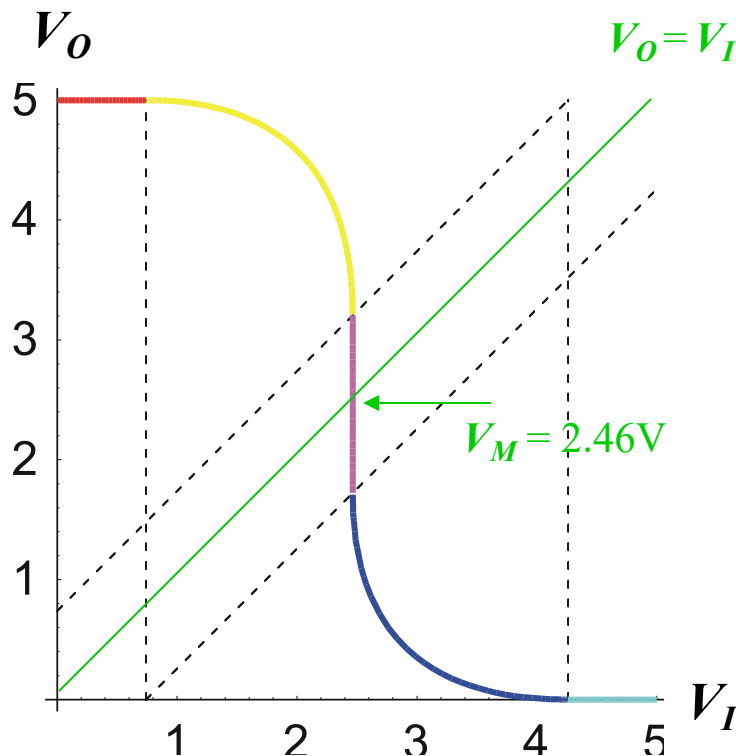
$$\frac{\partial V_O}{\partial V_I} = -1 \Rightarrow V_{IHm} = 2.89 \text{V} \quad V_{OLM} = 0.446 \text{V}$$

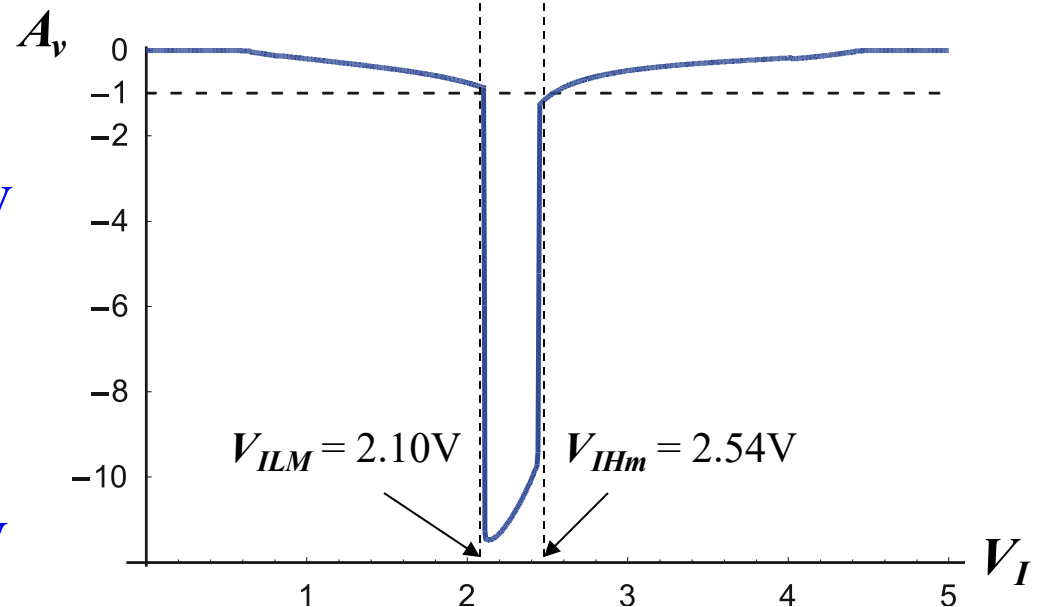
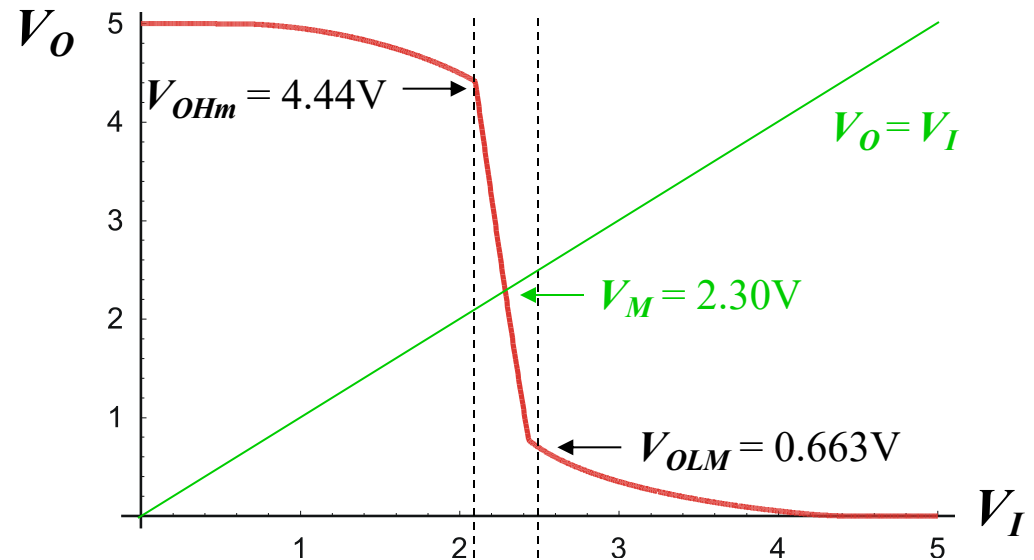
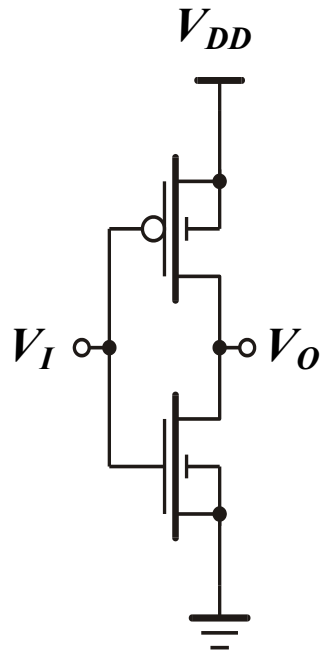
CALCOLO DI V_M

$$\begin{array}{ll} k_n' = 80 \mu\text{AV}^{-2} & k_p' = 27 \mu\text{AV}^{-2} \\ V_{Tn0} = 740 \text{mV} & V_{Tp0} = 740 \text{mV} \\ W_n = 1.80 \mu\text{m} & W_p = 5.33 \mu\text{m} \\ L_n = 1.20 \mu\text{m} & L_p = 1.20 \mu\text{m} \\ \lambda_n = 0 \text{V}^{-1} & \lambda_p = 0 \text{V}^{-1} \end{array}$$

$$V_M = V_{IS} \approx \frac{V_{Tn} + \sqrt{k_p / k_n} (V_{DD} - V_{Tp})}{1 + \sqrt{k_p / k_n}}$$

$$V_{Tn} = V_{Tp} \quad k_p / k_n = 1 \quad \rightarrow \quad V_M = \frac{V_{DD}}{2}$$





$$SW = V_{OHm} - V_{OLm} = V_{DD} = 5V$$

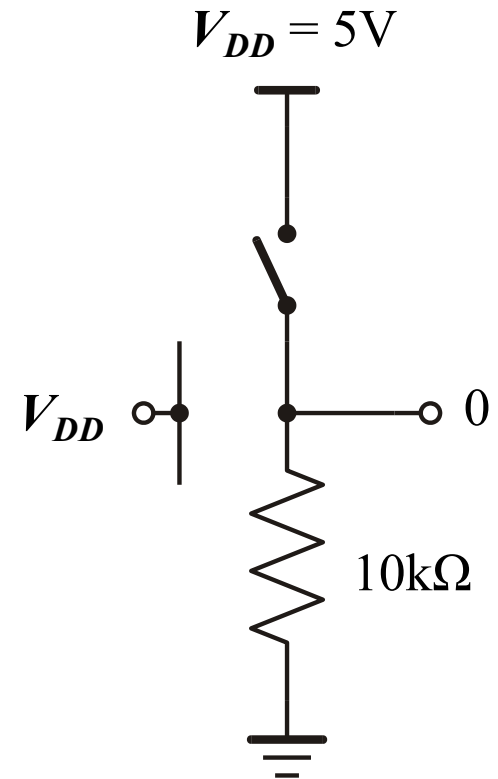
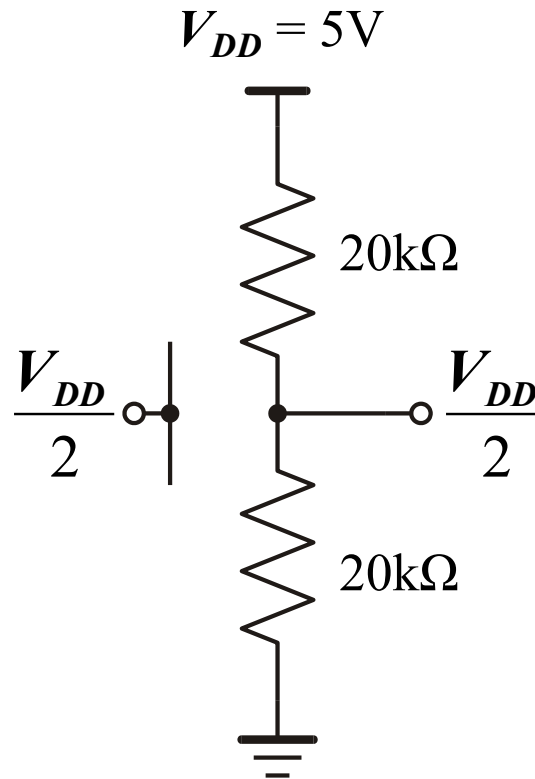
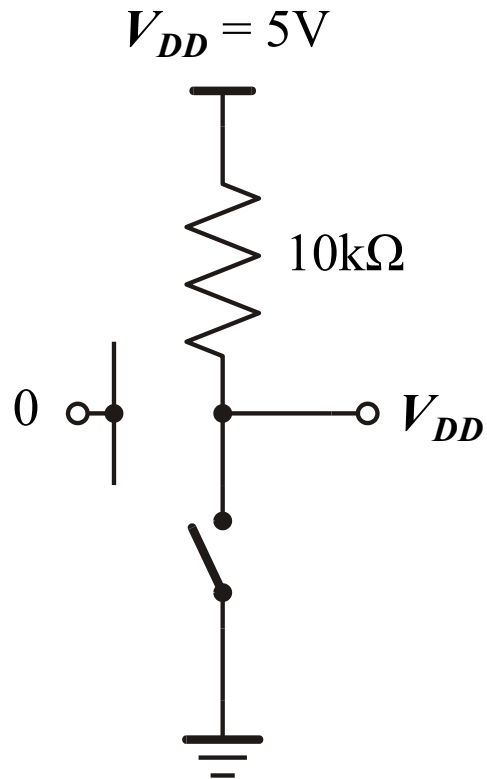
$$NM_H = V_{OHm} - V_{IHm} = 1.91V$$

$$NM_L = V_{ILm} - V_{OLm} = 1.44V$$

$$NM = \min\{NM_L, NM_H\} = 1.44V$$

EQUIVALENTE RESISTIVO

(AMI 1.2 μm)

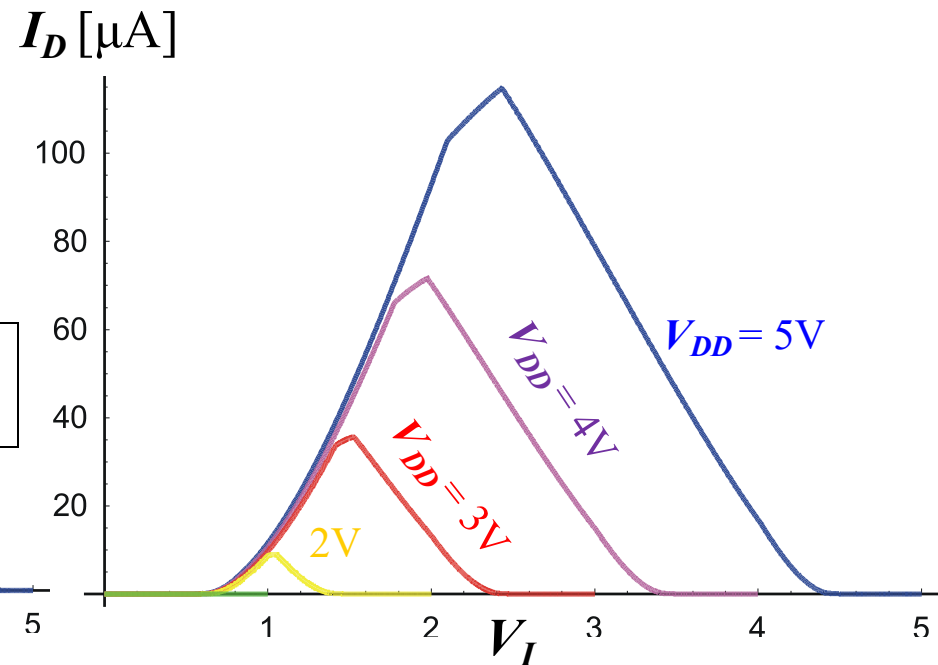
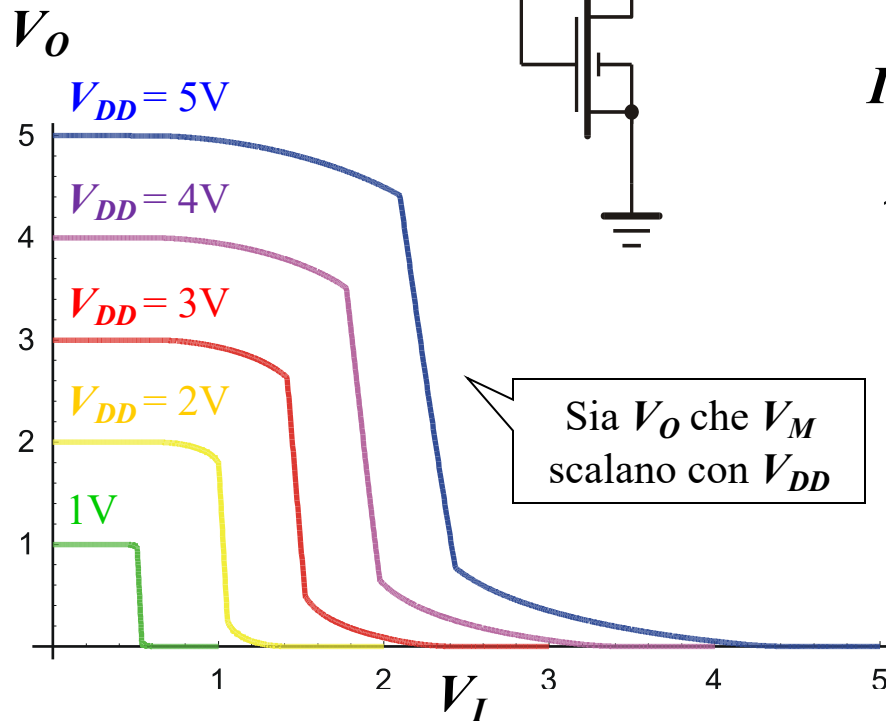
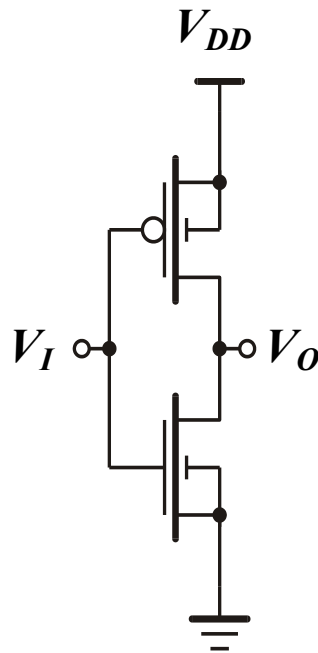


DIPENDENZA DA Vdd

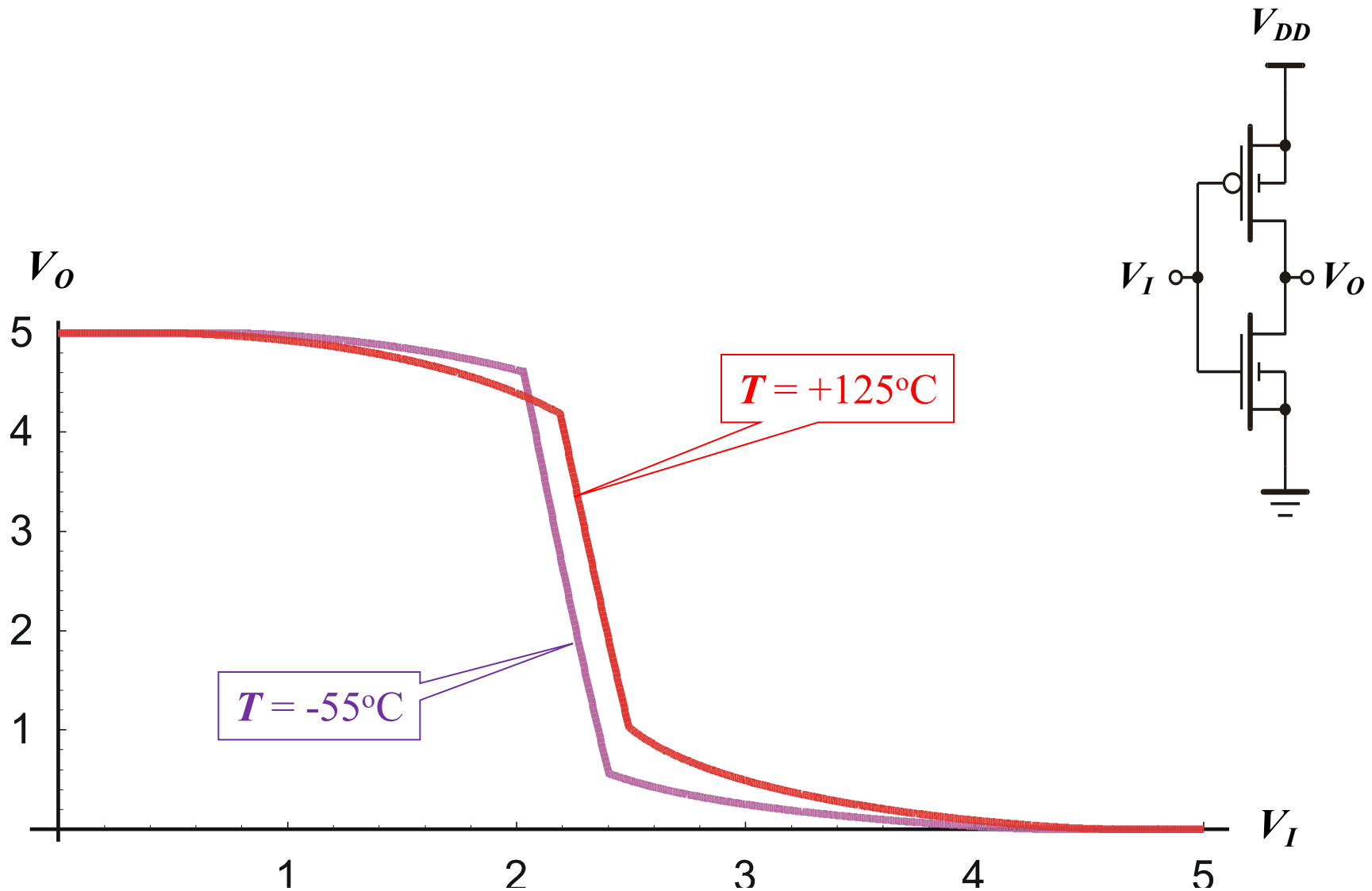
$$V_{Tn} = V_{Tp}$$

$$k_n = k_p$$

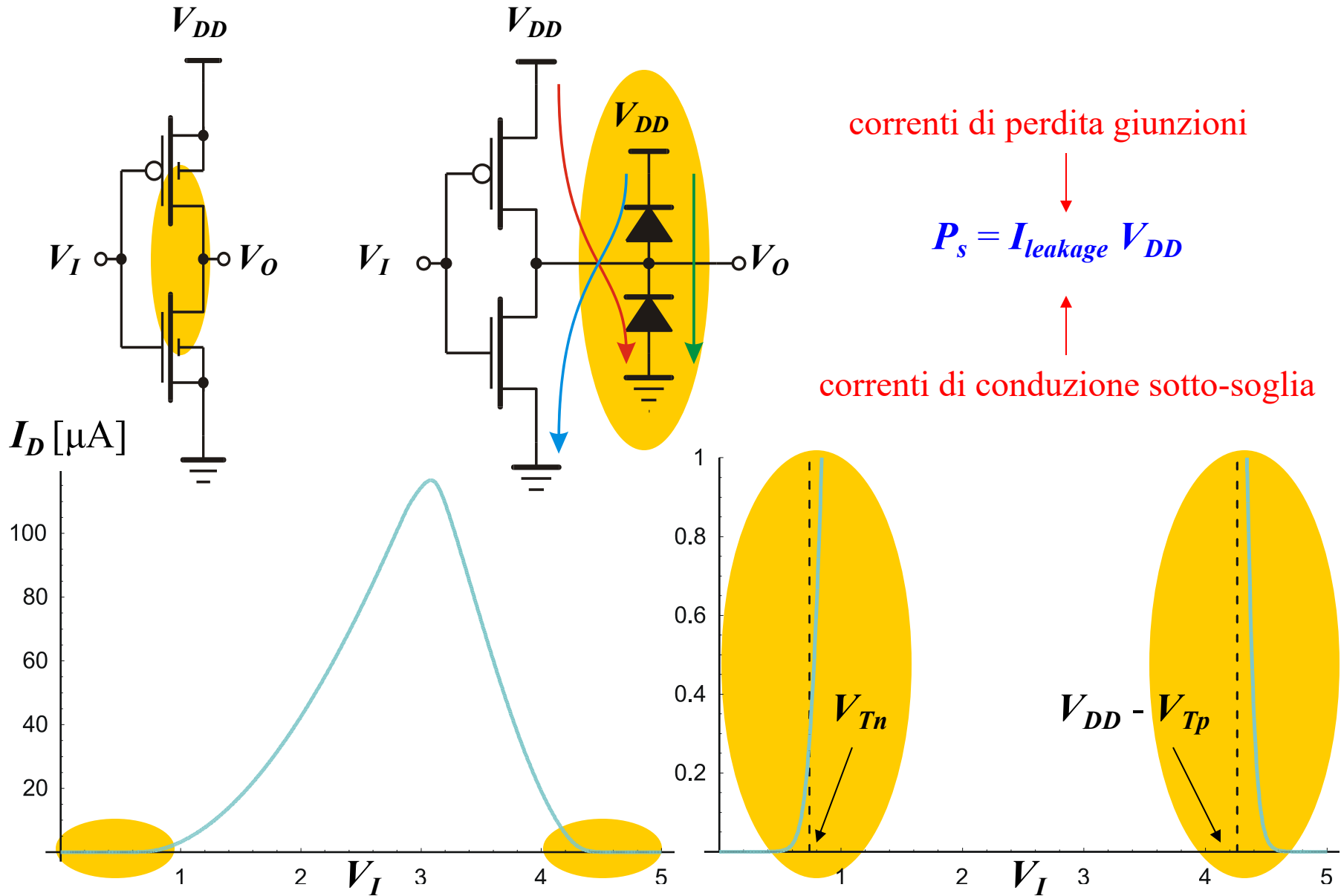
$$V_{IS} \approx \frac{V_{Tn} + \sqrt{k_p / k_n} (V_{DD} - V_{Tp})}{1 + \sqrt{k_p / k_n}} = \frac{V_{DD}}{2}$$

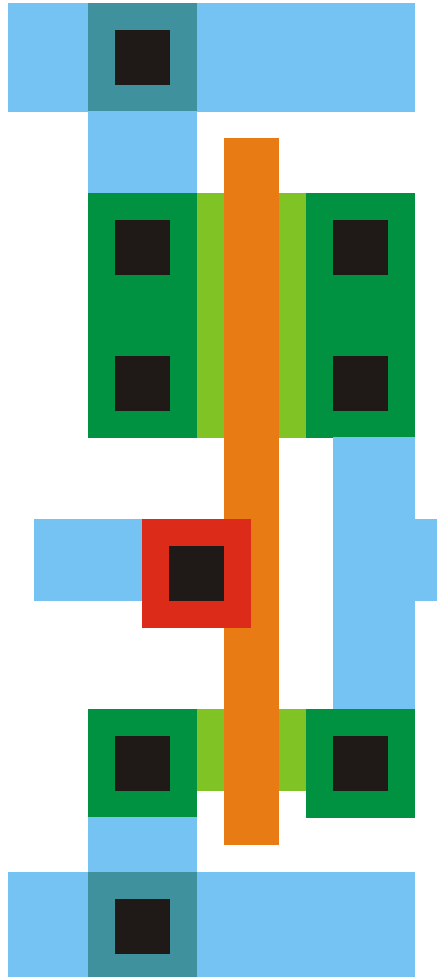


INFLUENZA DELLA TEMPERATURA

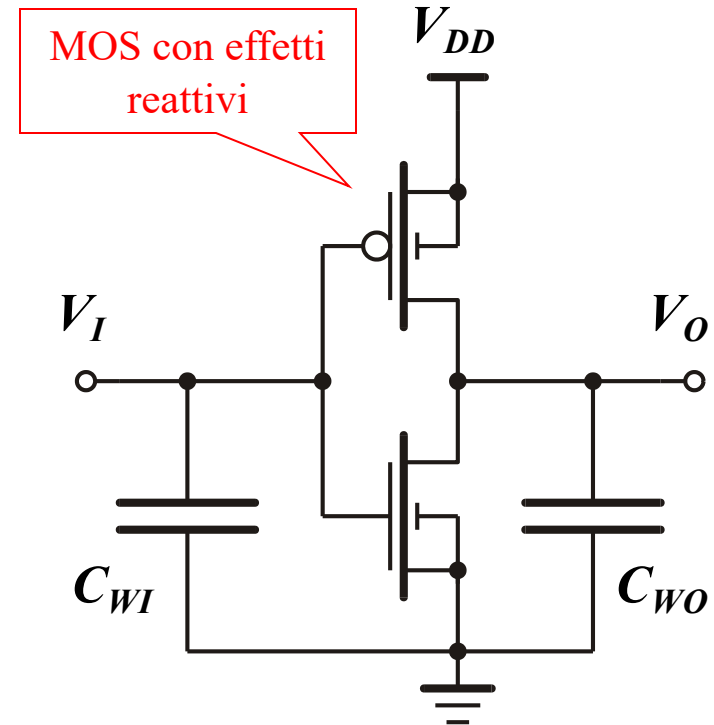


POTENZA STATICA DISSIPATA





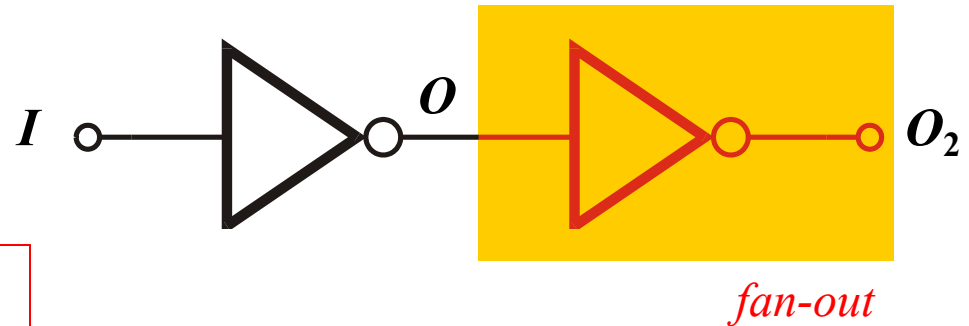
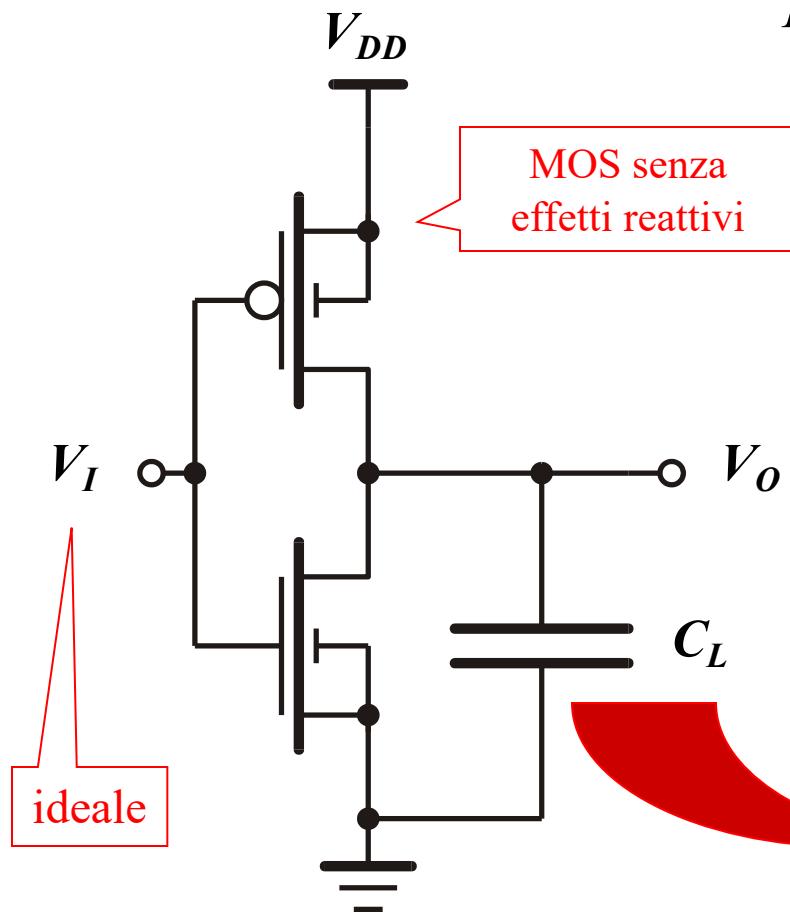
resistenze trascurate
< 5% del canale
dei MOS



- ✓ Tempi caratteristici
- ✓ Potenza dissipata
- ✓ Prodotto ritardo-consumo

CIRCUITO SEMPLIFICATO

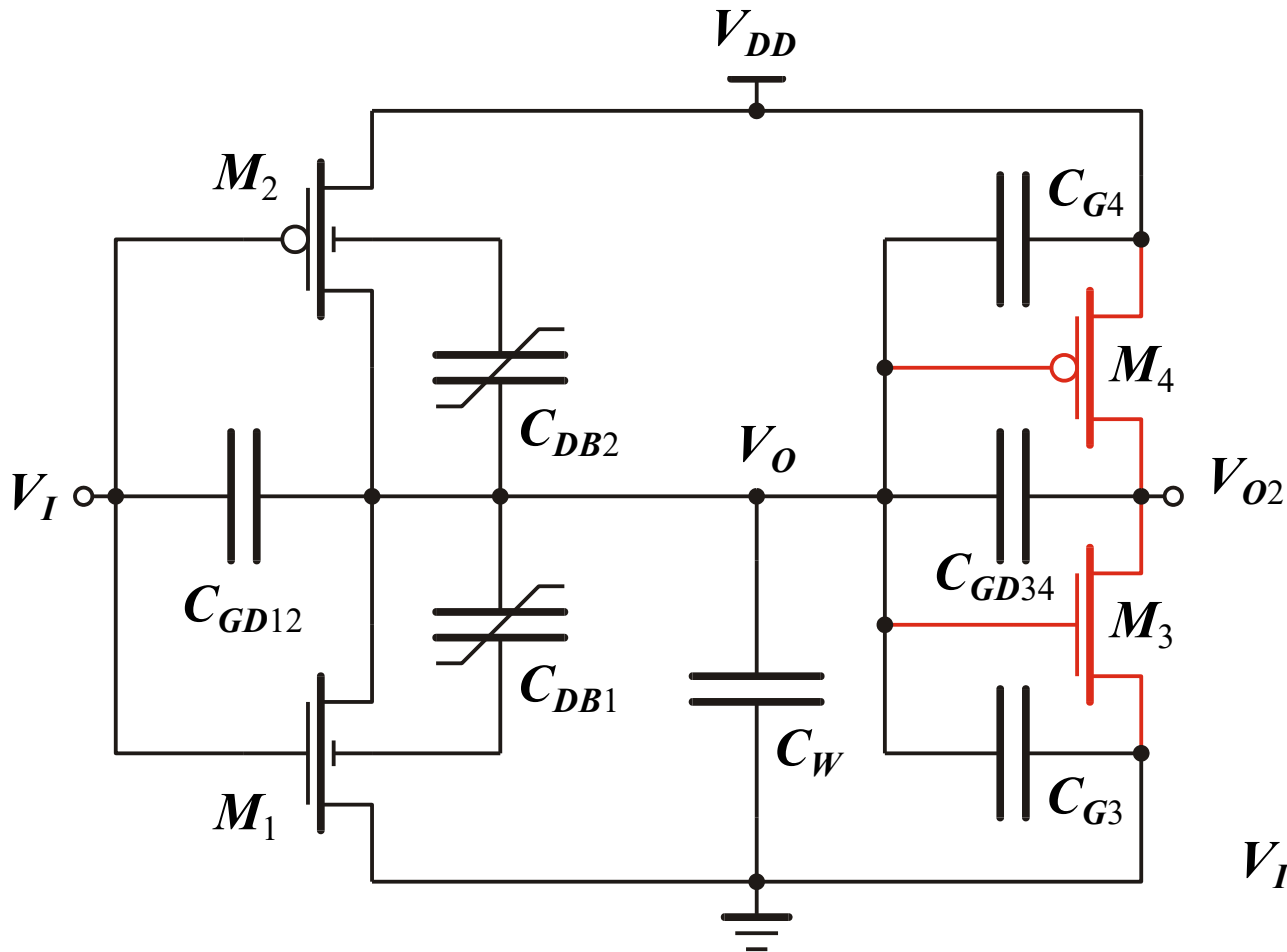
analisi manuale



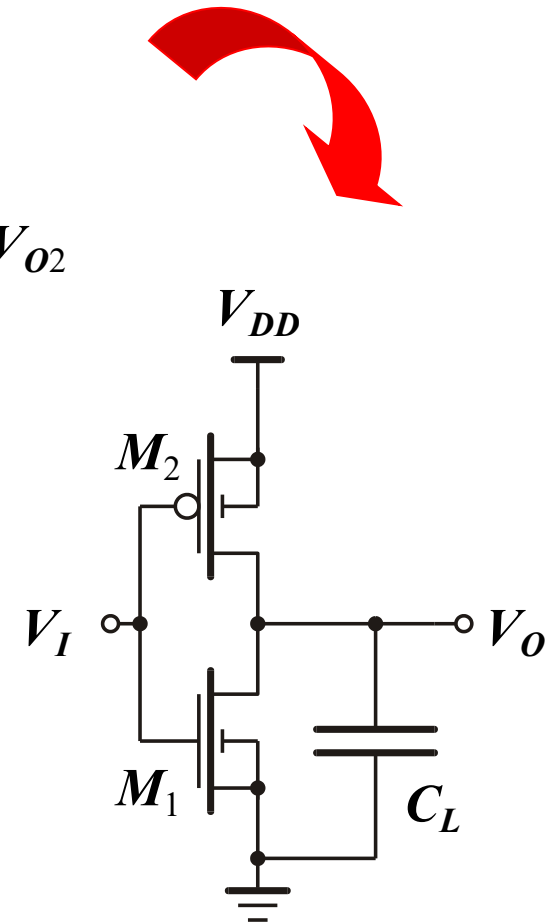
C_L tiene in conto:

- gli elementi capacitivi interni
- i collegamenti
- la porta (identica) a valle

CIRCUITO EQUIVALENTE DEI 2 NOT



Forte approssimazione!

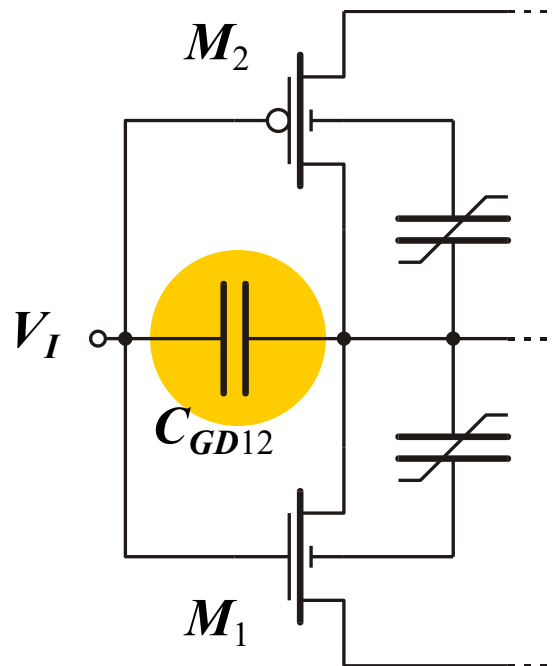


Trascurando la
dinamica di M_1 e M_2
in regione lineare

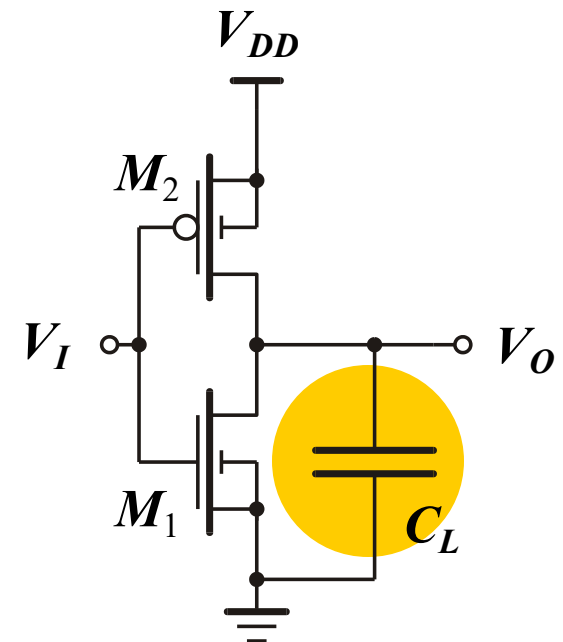
$$\rightarrow C_{GD12} = W_1 C_{on} + W_2 C_{op}$$

somma delle
capacità di *overlap*
di M_1 e M_2

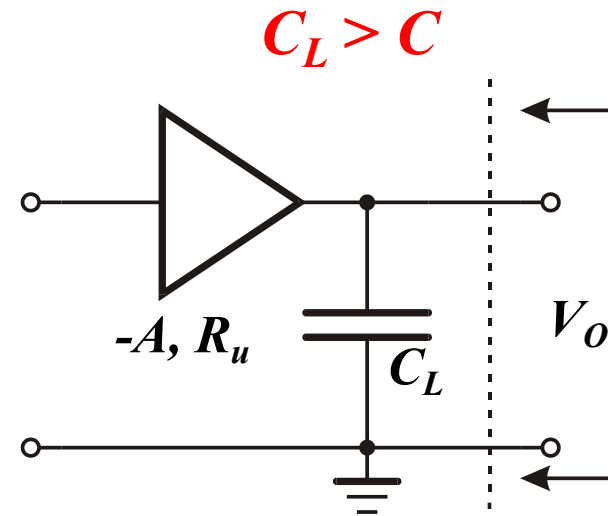
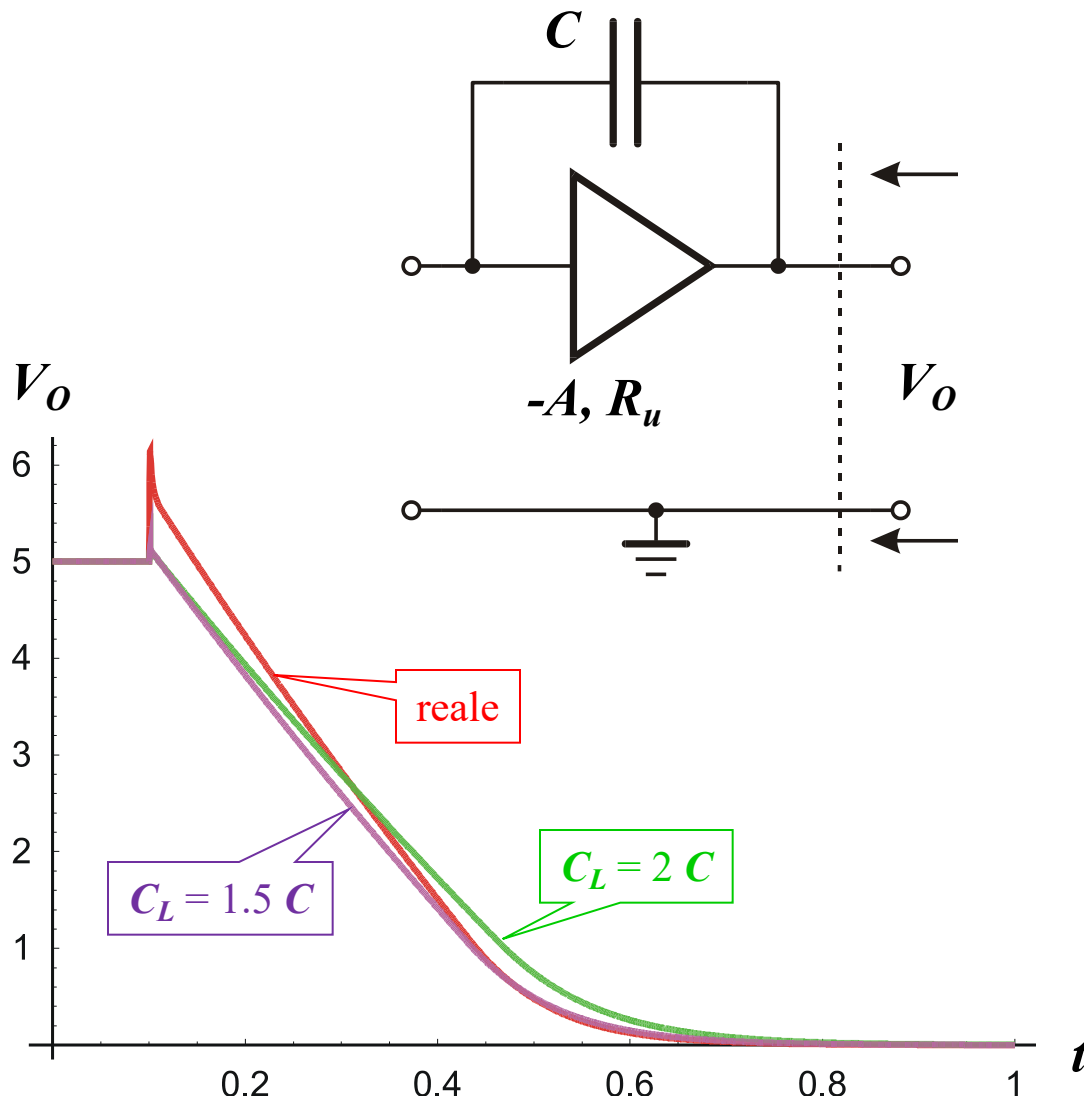
$C_o \rightarrow$ SPICE: CGDO



$$\Delta C_L = ?$$

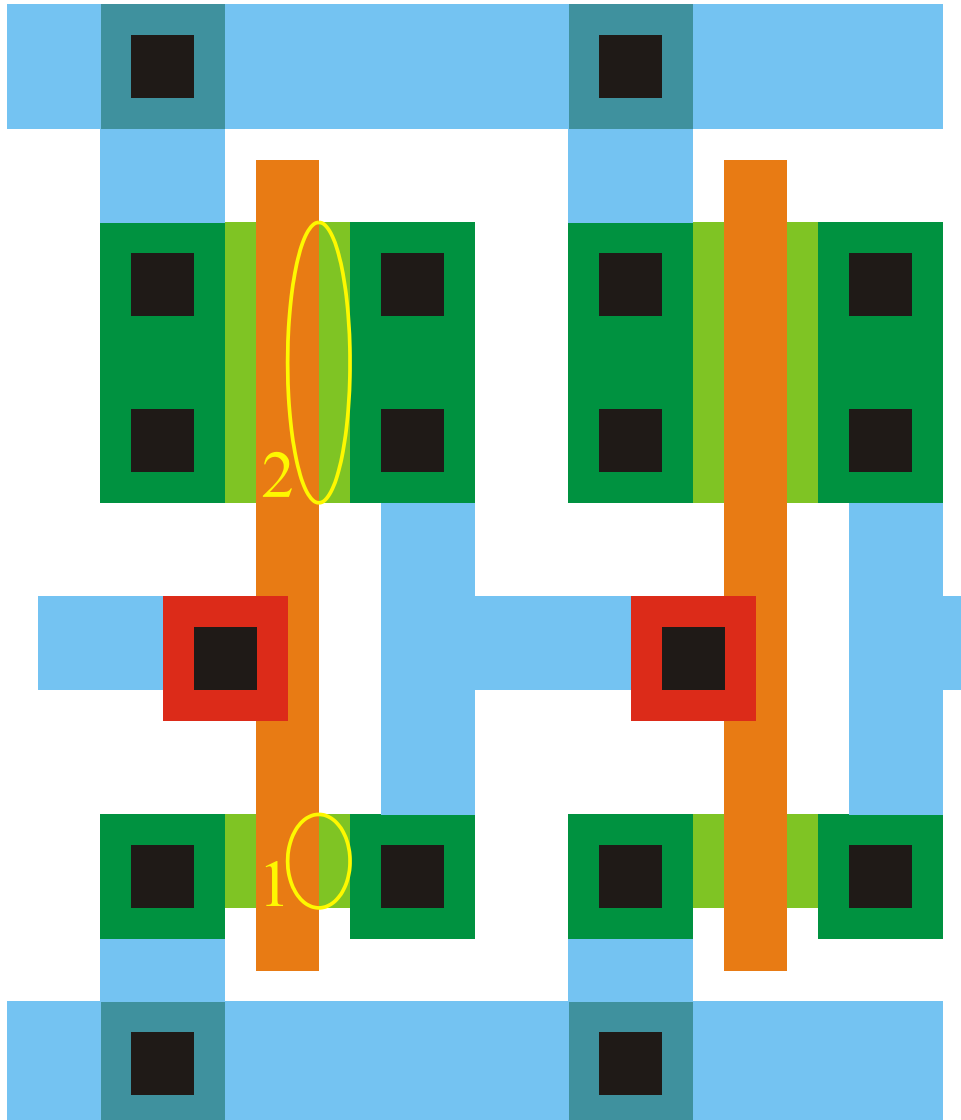


EFFETTO MILLER dall'uscita



$$V_o(t)|_{50\%} \Rightarrow C_L \approx 2C$$

$$V_o(t)|_{90\%} \Rightarrow C_L \approx 1.5C$$



$$C_{GD12} = W_1 C_{On} + W_2 C_{Op}$$

CMOS 1.2 μ m

$$C_{GDO_n} = 0.43 \text{ fF}/\mu\text{m}$$

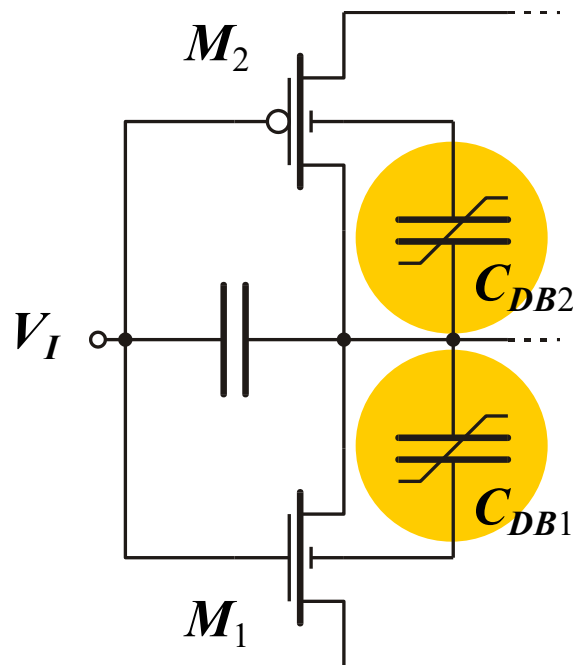
$$C_{GDO_p} = 0.43 \text{ fF}/\mu\text{m}$$

$$\begin{aligned} \Delta C_L &\approx 2 (W_1 + W_2) C_{GDO} \\ &= 6.19 \text{ fF} \end{aligned}$$

$$C_{DB} = C_{jD} A_D + C_{jswD} P_D$$

$$K_{eq} = -\frac{\Phi_B^m}{(V_h - V_l)(1-m)} \left[(\Phi_B - V_h)^{1-m} - (\Phi_B - V_l)^{1-m} \right]$$

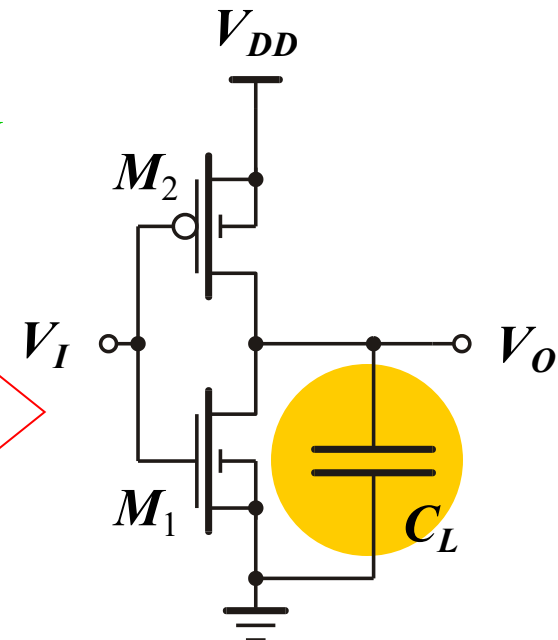
$$C_j = \frac{C_{j0}}{\left(1 - \frac{V}{\Phi_B}\right)^m}$$

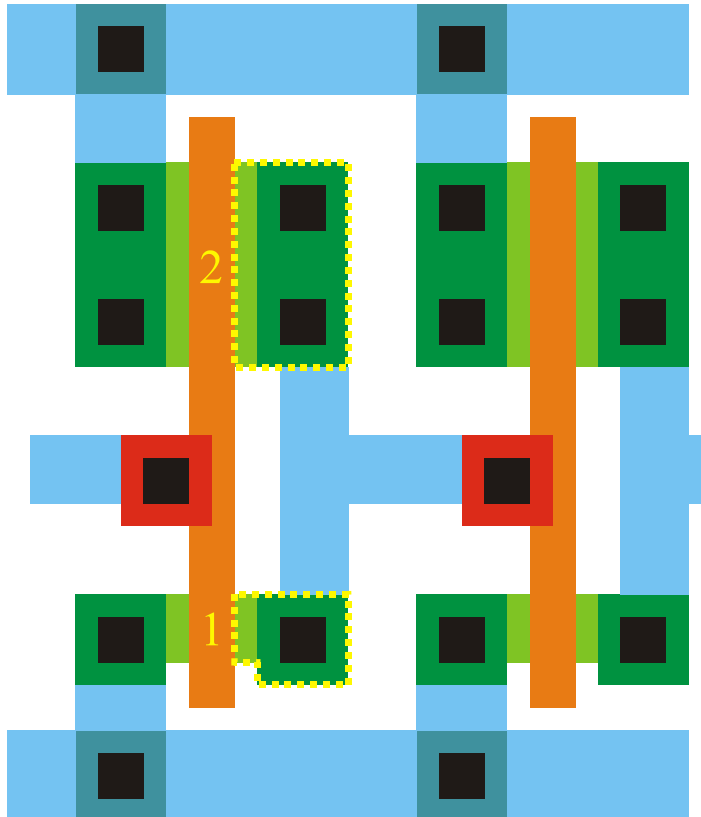


C_{jD0} → SPICE: CJ
 m → SPICE: MJ
 C_{jswD0} → SPICE: CJSW
 M_{sw} → SPICE: MJSW
 Φ_B → SPICE: PB

$$\Delta C_L \approx C_{DBeq1} + C_{DBeq2}$$

$$C_{eq} = K_{eq} C_{j0}$$





CMOS $1.2\mu\text{m}$ $\lambda = 0.6\mu\text{m}$
 $\Phi_{Bn} = 0.58\text{V}$ $\Phi_{Bp} = 0.64\text{V}$
 $C_{jD0n} = 0.3\text{mF}/\text{m}^2$ $C_{jD0p} = 0.5\text{mF}/\text{m}^2$
 $m_n = 0.66$ $m_p = 0.51$
 $C_{jswD0n} = 0.8\text{fF}/\mu\text{m}$ $C_{jswD0p} = 0.135\text{fF}/\mu\text{m}$
 $m_{swn} = 0.24$ $m_{swp} = 0.24$

$$C_{DBeq} = C_{jD0} K_{eqD} A_D + C_{jswD0} K_{eqDsw} P_D$$

$$K_{eq} = -\frac{\Phi_B^m}{(V_h - V_l)(1-m)} \left[(\Phi_B - V_h)^{1-m} - (\Phi_B - V_l)^{1-m} \right]$$

1

$$V_h = -5\text{V}$$

$$V_l = -2.5\text{V}$$

$$A_D = 19\lambda^2 = 6.84\mu\text{m}^2$$

$$P_D = 15\lambda = 9.0\mu\text{m}$$

$$K_{eqD} = 0.270$$

$$K_{eqDsw} = 0.620$$

2

$$V_h = -2.5\text{V}$$

$$V_l = 0\text{V}$$

$$A_D = 45\lambda^2 = 16.2\mu\text{m}^2$$

$$P_D = 19\lambda = 11.4\mu\text{m}$$

$$K_{eqD} = 0.617$$

$$K_{eqDsw} = 0.791$$

$$C_{DBeq1} = 5.02\text{fF}$$

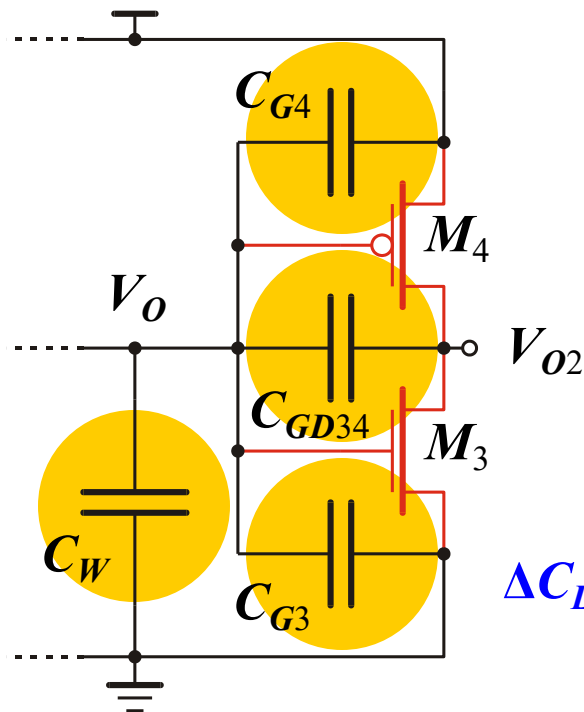
$$C_{DBeq2} = 6.21\text{fF}$$

$$\Delta C_L \approx C_{DBeq1} + C_{DBeq2} = 11.2\text{fF}$$

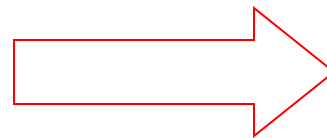
CONNESSIONE/SECONDO INVERTER

Hp: V_{O2} varia poco
nel transitorio di V_O

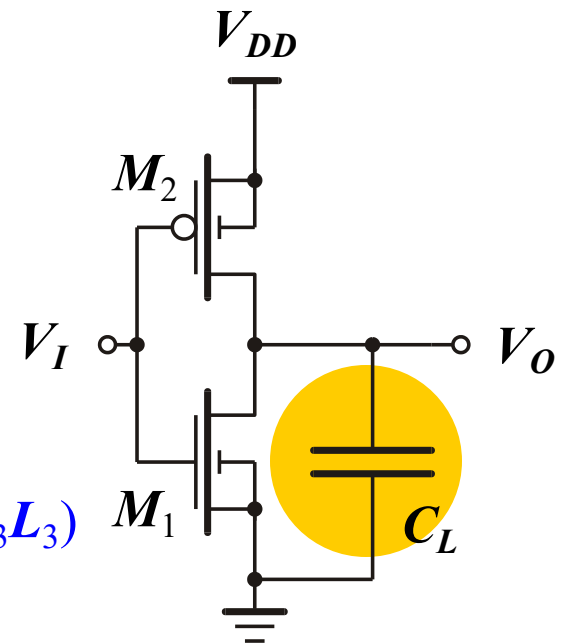
$$\longrightarrow C_{G4} + C_{GD34} + C_{G3} \approx C_{ox}(W_4L_4 + W_3L_3)$$

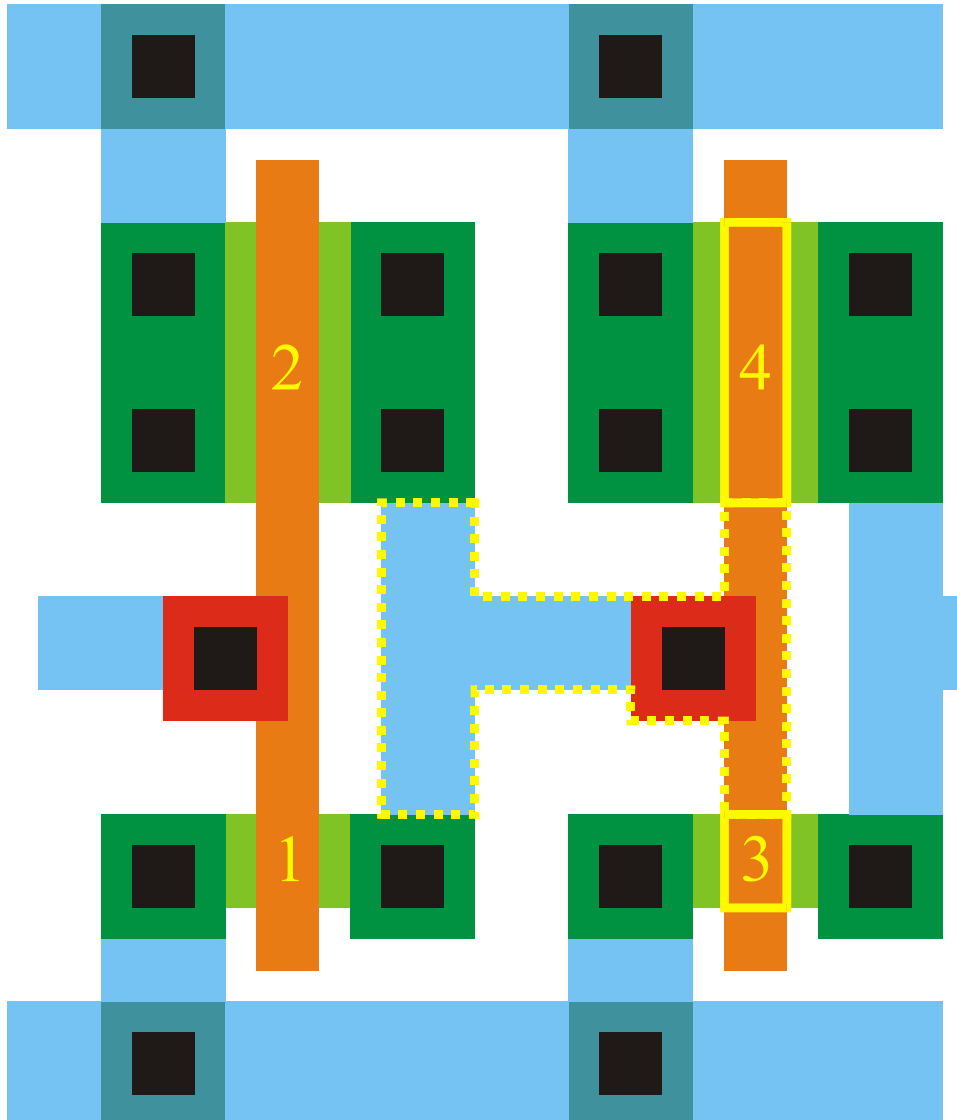


C_W : estratto dal *layout*



$$\Delta C_L \approx C_W + C_{ox}(W_4L_4 + W_3L_3)$$





$$\Delta C_L \approx C_W + C_{ox}(W_4 L_4 + W_3 L_3)$$

(AMI 1.2 μ m 1998)

$$\lambda = 0.6\mu\text{m}$$

$$C_{Mf} = 44\text{aF}/\mu\text{m}$$

$$C_M = 20\text{aF}/\mu\text{m}^2$$

$$C_P = 36\text{aF}/\mu\text{m}^2$$

$$C_W = C_M A_M + C_P A_P + C_{Mf} P_M$$

$$A_M = 45\lambda^2 \quad A_P = 32\lambda^2 \quad P_M = 34\lambda$$

$$C_W = 1.64\text{fF}$$

CMOS 1.2 μ m

$$t_{ox} = 20\text{nm}$$

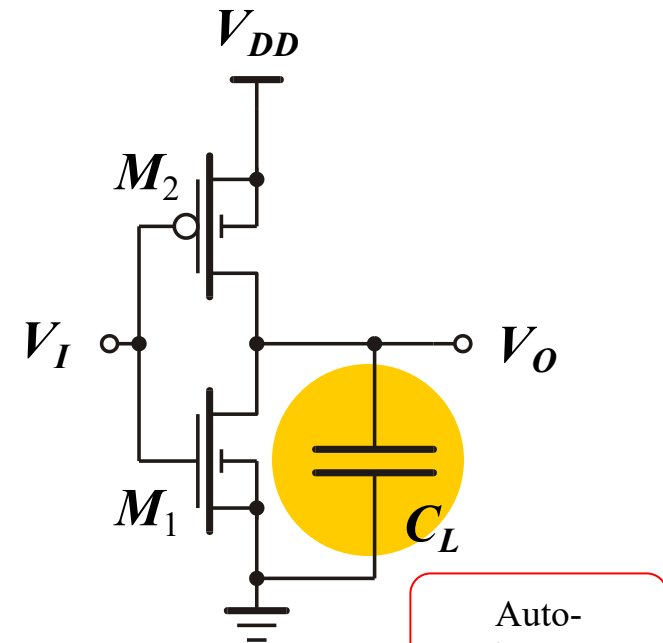
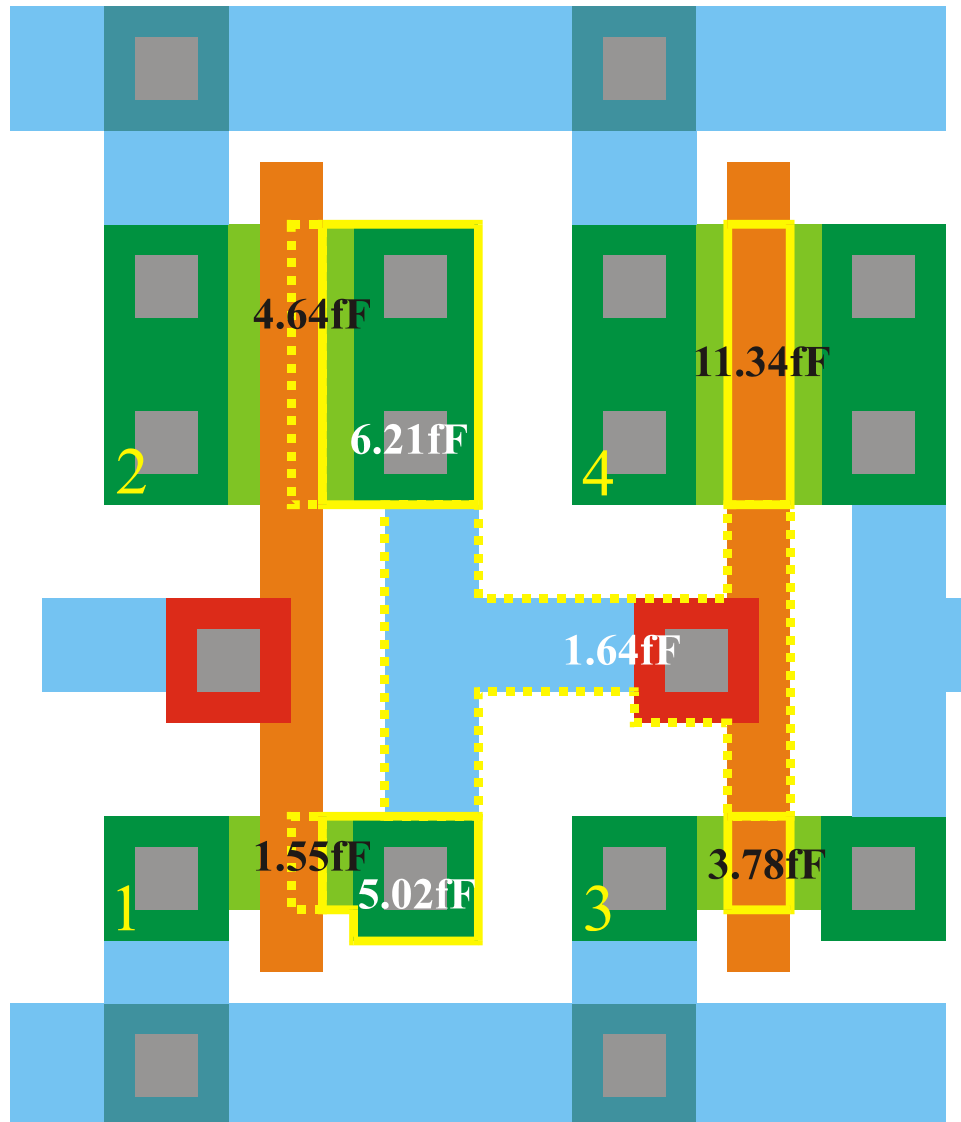
$$\epsilon_{ox} = 3.5 \times 10^{-13}\text{F}/\text{cm}$$

$$C_{ox}(W_4 L_4 + W_3 L_3) =$$

$$(9 \times 2 + 3 \times 2) \lambda^2 \epsilon_{ox} / t_{ox} =$$

$$15.12\text{fF}$$

$$\Delta C_L \approx 16.8\text{fF}$$



Auto-caricamento

capacità intrinseca

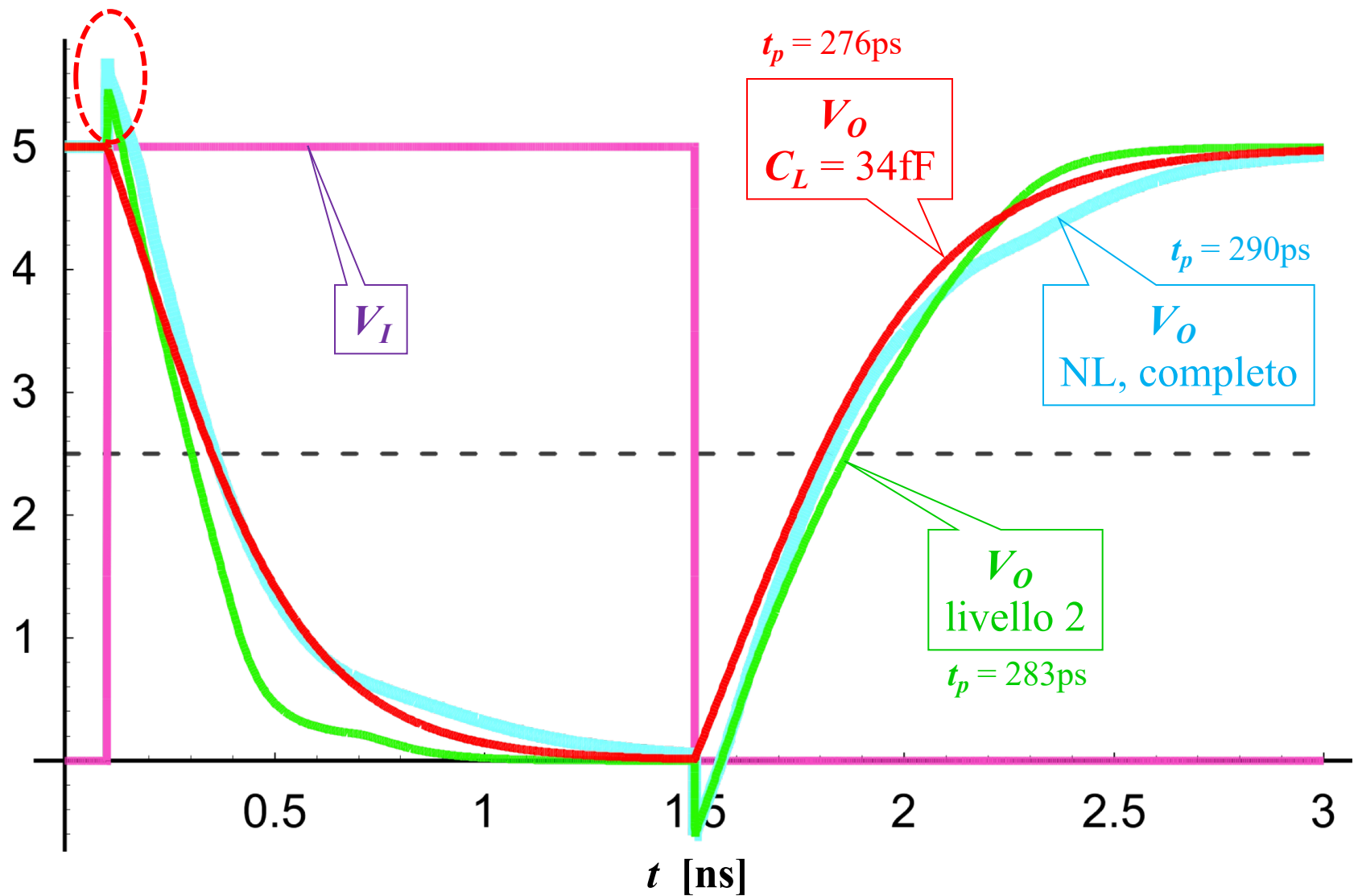
$$C_L = 2 C_{GD12} + C_{DBeq1} + C_{DBeq2} + C_W + C_{ox}(W_4 L_4 + W_3 L_3) =$$

34.2fF

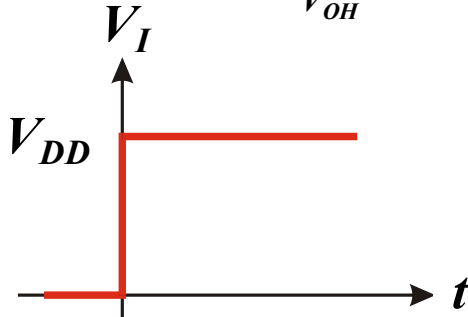
capacità estrinseca
(cresce circa linearmente
con il *fan-out*)

Calcolo di C_L per la determinazione di t_p

		H → L	L → H
		50%	50%
$2(C_{GD1} + C_{GD2})$	$2(W_1 + W_2)CGDO$	6.2fF	6.2fF
C_{DBeq1}	$K_{eqDn}A_{Dn}CJ_n + K_{eqDsw n}P_{Dn}CJSW_n$	5.0fF	6.7fF
C_{DBeq2}	$K_{eqDp}A_{Dp}CJ_p + K_{eqDsw p}P_{Dp}CJSW_p$	6.2fF	4.0fF
C_W	estrazione dal <i>LAYOUT</i>	1.6fF	1.6fF
$C_{G3} + C_{G4}$	$C_{ox}(W_3L_3 + W_4L_4)$	15.1fF	15.1fF
C_L	Σ	34.2fF	33.7fF

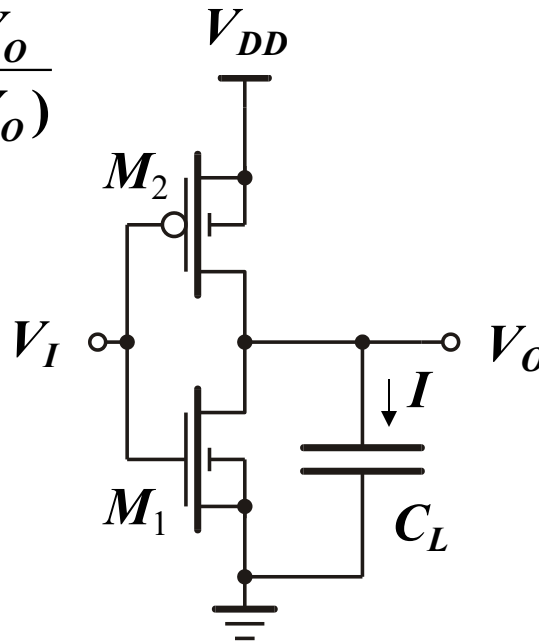


$$t_{pHL} = C_L \int_{V_{OH}}^{(V_{OH}+V_{OL})/2} \frac{dV_o}{I(V_o)}$$



$$V_{GS1} - V_{Tn} < V_{DS1}$$

$$V_I - V_{Tn} < V_O$$



$$t = 0^-: V_I = 0V$$

$$V_O = V_{DD}$$

$$Q_{CL} = V_{DD} C_L$$

$$M_1 \text{ off}, M_2 \text{ lin}$$

$$t = 0^+: V_I = V_{DD}$$

$$V_O = V_{DD}$$

$$Q_{CL} = V_{DD} C_L$$

$$M_1 \text{ sat}, M_2 \text{ off}$$

$$\rightarrow M_1 \text{ saturo se } V_O > V_{DD} - V_{Tn} : t \in [0, t_s]$$

$$M_1 \text{ lineare se } V_{DD} - V_{Tn} > V_O > V_{DD}/2 : t \in [t_s, t_{pHL}]$$

M_1 saturo:

$$I_{D1} = \frac{k_n}{2} (V_{DD} - V_{Tn})^2 = -I = -C_L \frac{dV_o}{dt} \quad t_s = \frac{2V_{Tn}C_L}{k_n(V_{DD} - V_{Tn})^2} = 71.5\text{ps}$$

$$V_{DD} = 5V \quad C_L = 34.2\text{fF} \quad V_{Tn} = 0.743V \quad k'_n = 19.6\mu\text{A/V}^2 \quad L_1 = 1.2\mu\text{m} \quad W_1 = 1.8\mu\text{m} \quad L_D = 0.15\mu\text{m}$$

M_1 saturo se $V_O > V_{DD} - V_{Tn}$: $t \in [0, t_s]$

M_1 lineare se $V_{DD} - V_{Tn} > V_O > V_{DD}/2$: $t \in [t_s, t_{pHL}]$

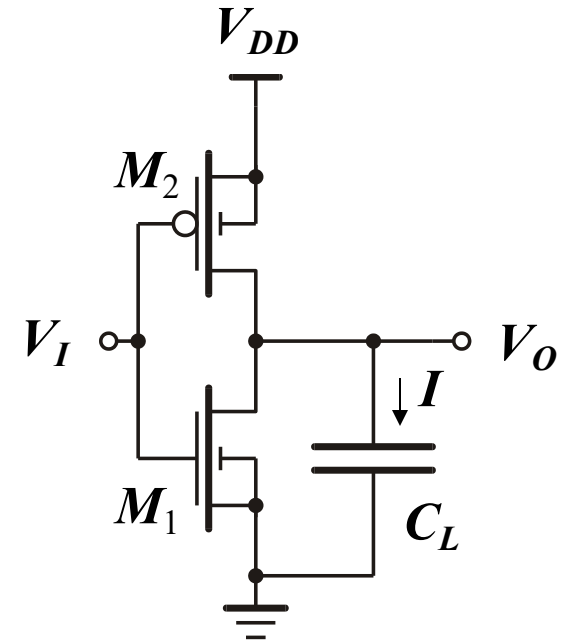
M_1 lineare:

$$I_{D1} = k_n \left[(V_{DD} - V_{Tn})V_O - \frac{V_O^2}{2} \right] = -I = -C_L \frac{dV_O}{dt}$$

$$t_{pHL} - t_s = -\frac{2C_L}{k_n} \int_{V_{DD}-V_{Tn}}^{V_{DD}/2} \frac{dV}{2V_O(V_{DD} - V_{Tn}) - V_O^2}$$

$$t_{pHL} - t_s = \frac{C_L}{k_n(V_{DD} - V_{Tn})} \ln \frac{3V_{DD} - 4V_{Tn}}{V_{DD}} = 179.9\text{ps}$$

$$t_{pHL} = \frac{C_L}{k_n(V_{DD} - V_{Tn})} \left[\frac{2V_{Tn}}{V_{DD} - V_{Tn}} + \ln \frac{3V_{DD} - 4V_{Tn}}{V_{DD}} \right] = 251.4\text{ps}$$

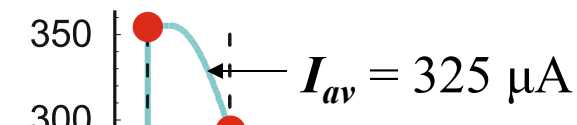


$$V_{DD} = 5\text{V} \quad C_L = 34.2\text{fF} \quad V_{Tn} = 0.743\text{V} \quad k'_n = 19.6\mu\text{A/V}^2 \quad L_1 = 1.2\mu\text{m} \quad W_1 = 1.8\mu\text{m} \quad L_D = 0.15\mu\text{m}$$

SEMPLIFICAZIONE(1)

$$C_L \frac{dV_o}{dt} = I(V_o) \quad t_{pHL} = C_L \int_{V_{OH}}^{(V_{OH}+V_{OL})/2} \frac{dV_o}{I(V_o)}$$

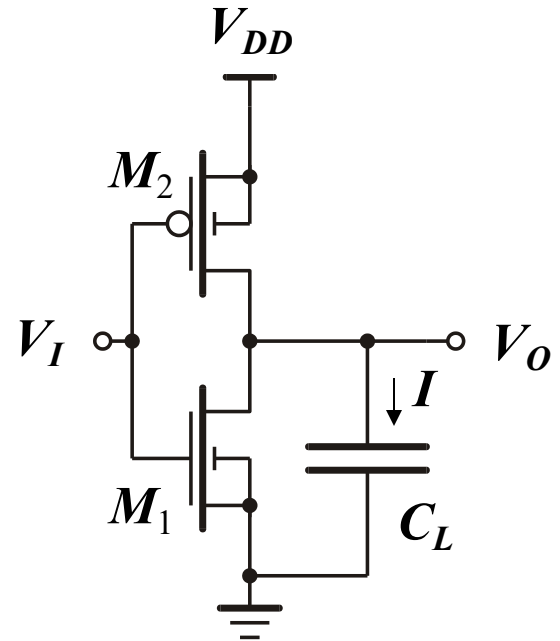
$$I(V_o) \leftarrow I_{av} = \frac{I(V_{DD}) + I(V_{DD}/2)}{2}$$



$$t_{pHL} \approx C_L \frac{V_{DD}/2 - V_{DD}}{I_{av}} = -C_L \frac{V_{DD}}{2I_{av}}$$

$$I_{av} = -\frac{1}{2} \left\{ \frac{k_n}{2} (V_{DD} - V_{Tn})^2 + k_n \left[(V_{DD} - V_{Tn}) \frac{V_{DD}}{2} - \frac{V_{DD}^2}{8} \right] \right\}$$

$$t_{pHL} \approx -\frac{C_L V_{DD}}{2I_{av}} = 263 \text{ ps}$$



$$V_{DD} = 5V \quad C_L = 34.2 \text{ fF} \quad V_{Tn} = 0.743V \quad k'_n = 19.6 \mu\text{A/V}^2 \quad L_1 = 1.2 \mu\text{m} \quad W_1 = 1.8 \mu\text{m} \quad L_D = 0.15 \mu\text{m}$$

SEMPLIFICAZIONE(2)

$$C_L \frac{dV_o}{dt} = I(V_o) \quad t_{pHL} = C_L \int_{V_{OH}}^{(V_{OH}+V_{OL})/2} \frac{dV_o}{I(V_o)}$$

$$I(V_o) \leftarrow I_{sat} = I(V_{DD})$$

$$I_{sat} = -\frac{k_n}{2} (V_{DD} - V_{Tn})^2$$

$$t_{pHL} \approx -C_L \frac{V_{DD}}{2I_{sat}} = \frac{C_L V_{DD}}{k_n (V_{DD} - V_{Tn})^2} = 241 \text{ ps}$$

$$V_{Tn} \ll V_{DD} \Rightarrow t_{pHL} \approx \frac{C_L}{V_{DD} k_n} = 174 \text{ ps}$$

$I_{D1} [\mu A]$

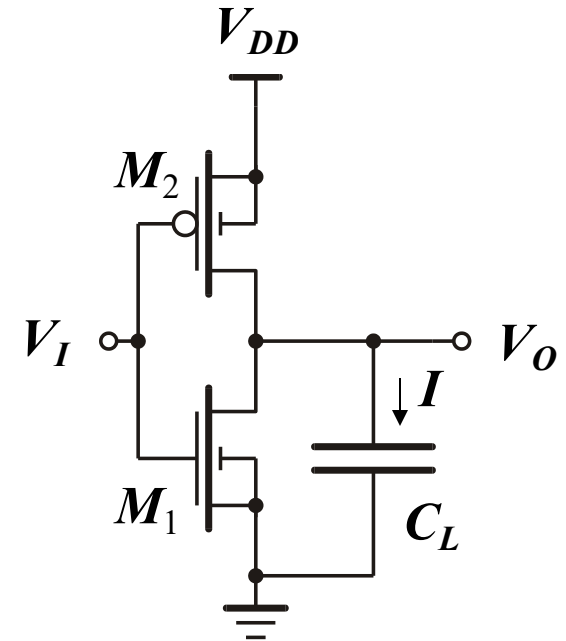
$I_{sat} = 355 \mu A$

350
300
250
200
150
100
50

0.5 1 1.5 2 2.5 3

$t [\text{ns}]$

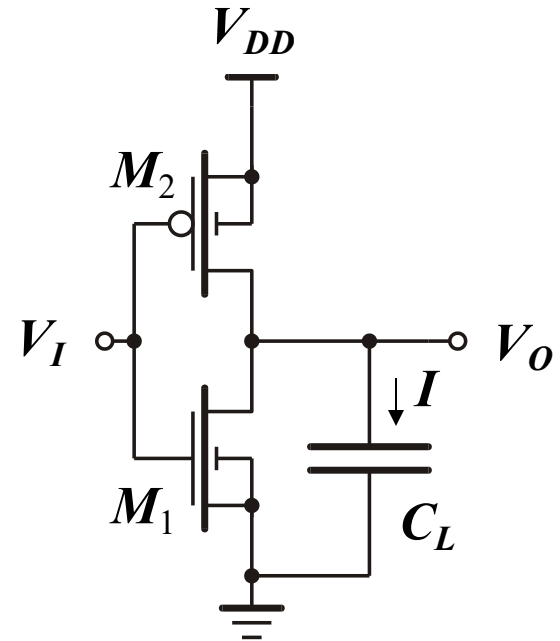
$V_{DD} = 5V$ $C_L = 34.2 \text{ fF}$ $V_{Tn} = 0.743V$ $k'_n = 19.6 \mu A/V^2$ $L_1 = 1.2 \mu m$ $W_1 = 1.8 \mu m$ $L_D = 0.15 \mu m$



CONSIDERAZIONI SUL TP

$$t_{pHL} \approx \frac{C_L}{V_{DD} k_n} = 174\text{ps} \quad (256\text{ps}) \quad t_{pLH} \approx \frac{C_L}{V_{DD} k_p} = 211\text{ps} \quad (323\text{ps})$$

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} \approx \frac{C_L}{2V_{DD}} \left(\frac{1}{k_n} + \frac{1}{k_p} \right) = 193\text{ps} \quad (290\text{ps})$$



Per ridurre il tempo di propagazioni, bisogna

- ✓ ridurre C_L : ovvero, 1) le diffusioni interne del *gate*; 2) le interconnessioni; 3) il *fan-out*
- ✓ aumentare V_{DD} : comporta aumento di P_d (con il quadrato!) e aumento del campo elettrico
- ✓ aumentare k_p e k_n : ovvero W , ma questo aumenta C_L (capacità intrinseca propria e a monte, ma non C_W)

$$V_{DD} = 5\text{V} \quad C_L = 34.2\text{fF} \quad k'_n = 19.6\mu\text{A/V}^2 \quad k'_p = 5.4\mu\text{A/V}^2 \quad L_{eff} = 0.9\mu\text{m} \quad W_1 = 1.8\mu\text{m} \quad W_2 = 5.4\mu\text{m}$$

OTTIMIZZAZIONE DI t_p

$L_n = L_p = 2\lambda$ (min) $\leftarrow$ Vincolo di area minima

$k_n = k_p \left[\frac{k_n'}{k_p'} = \frac{\mu_n}{\mu_p} \right] \Rightarrow W_p = \frac{\mu_n}{\mu_p} W_n \leftarrow NM$ ottimo

Rinunciando a NM ottimo:

$W_p \downarrow \Rightarrow C_L \downarrow \Rightarrow t_{pHL} \downarrow \quad t_{pLH} \uparrow \Rightarrow W_p$ ottimo?

$$C_L \approx (C_{D1} + C_{D2}) + (C_{G3} + C_{G4}) + C_W$$

$$\alpha = \frac{W_p / L_p}{W_n / L_n} = \frac{W_p}{W_n}$$

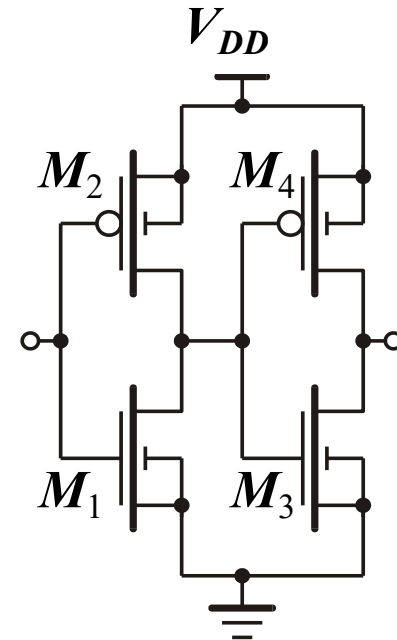
$$\begin{cases} C_L \approx (1+\alpha)(C_{D1} + C_{G3}) + C_W \\ t_p \approx \frac{C_L}{2 V_{DD}} \left(\frac{1}{k_n} + \frac{1}{k_p} \right) = \frac{C_L}{2 V_{DD} k_n} \left(1 + \frac{\mu_n}{\mu_p} \alpha \right) \end{cases} \rightarrow \frac{\partial t_p}{\partial \alpha} = 0$$

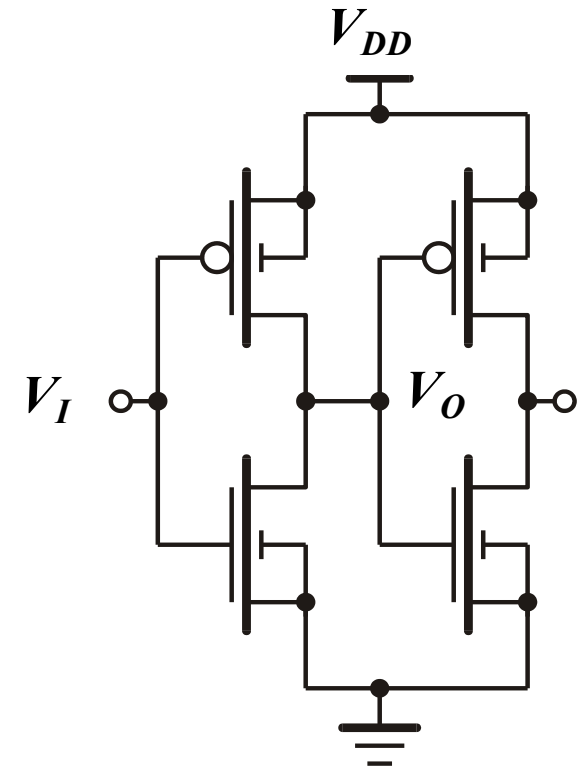
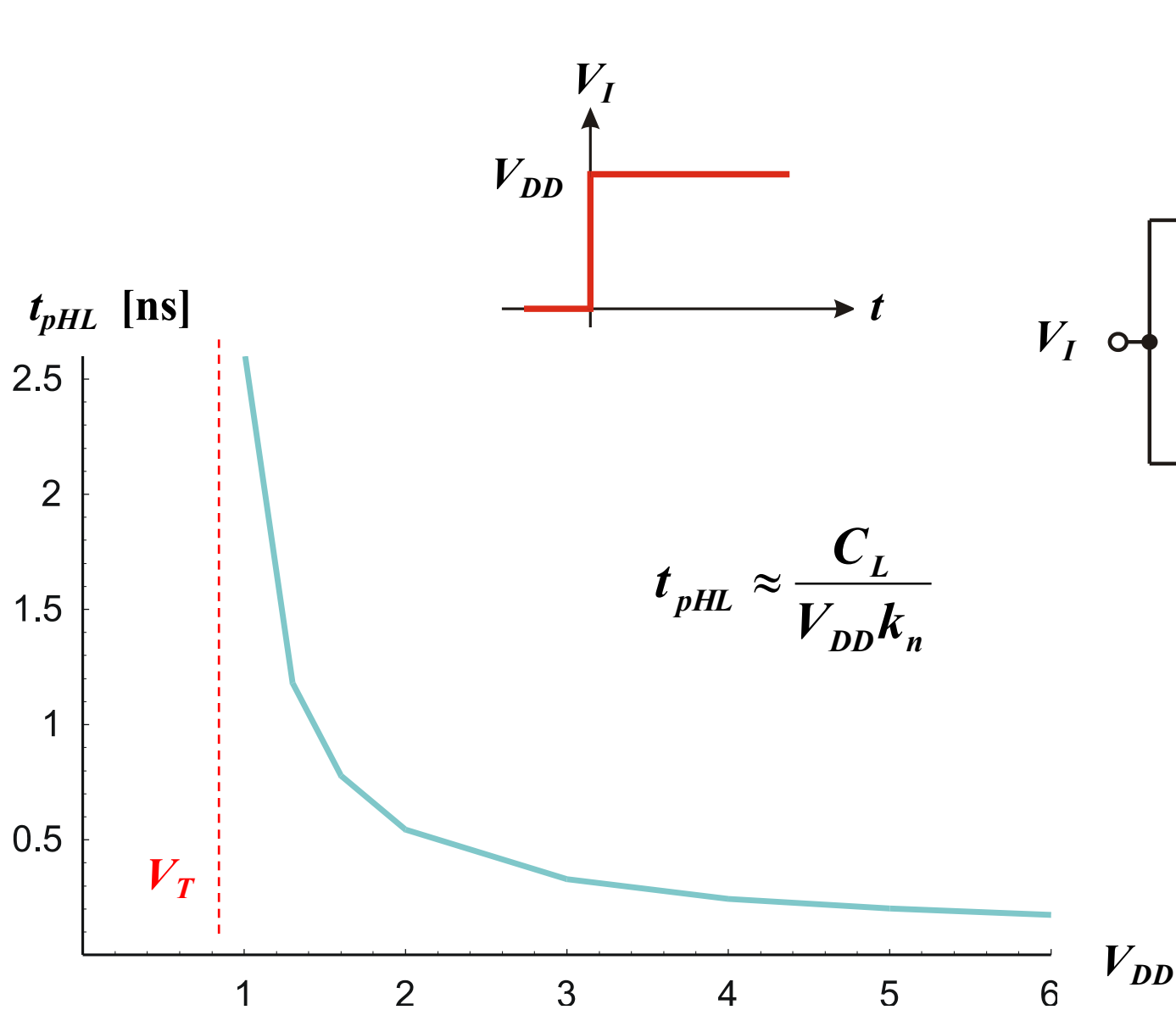
Esempio: $\mu_n / \mu_p = 3$

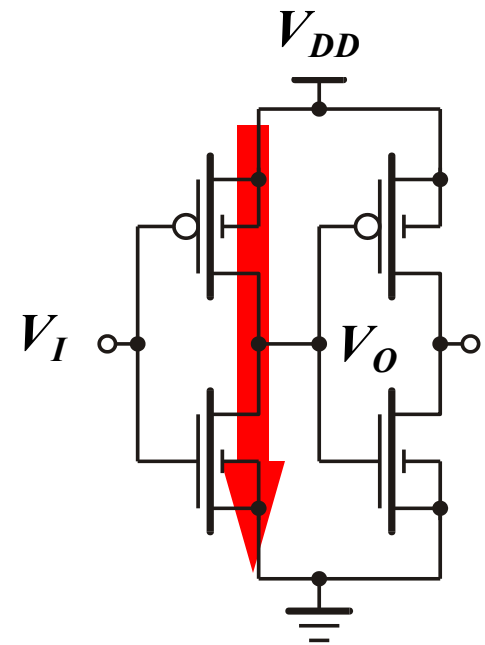
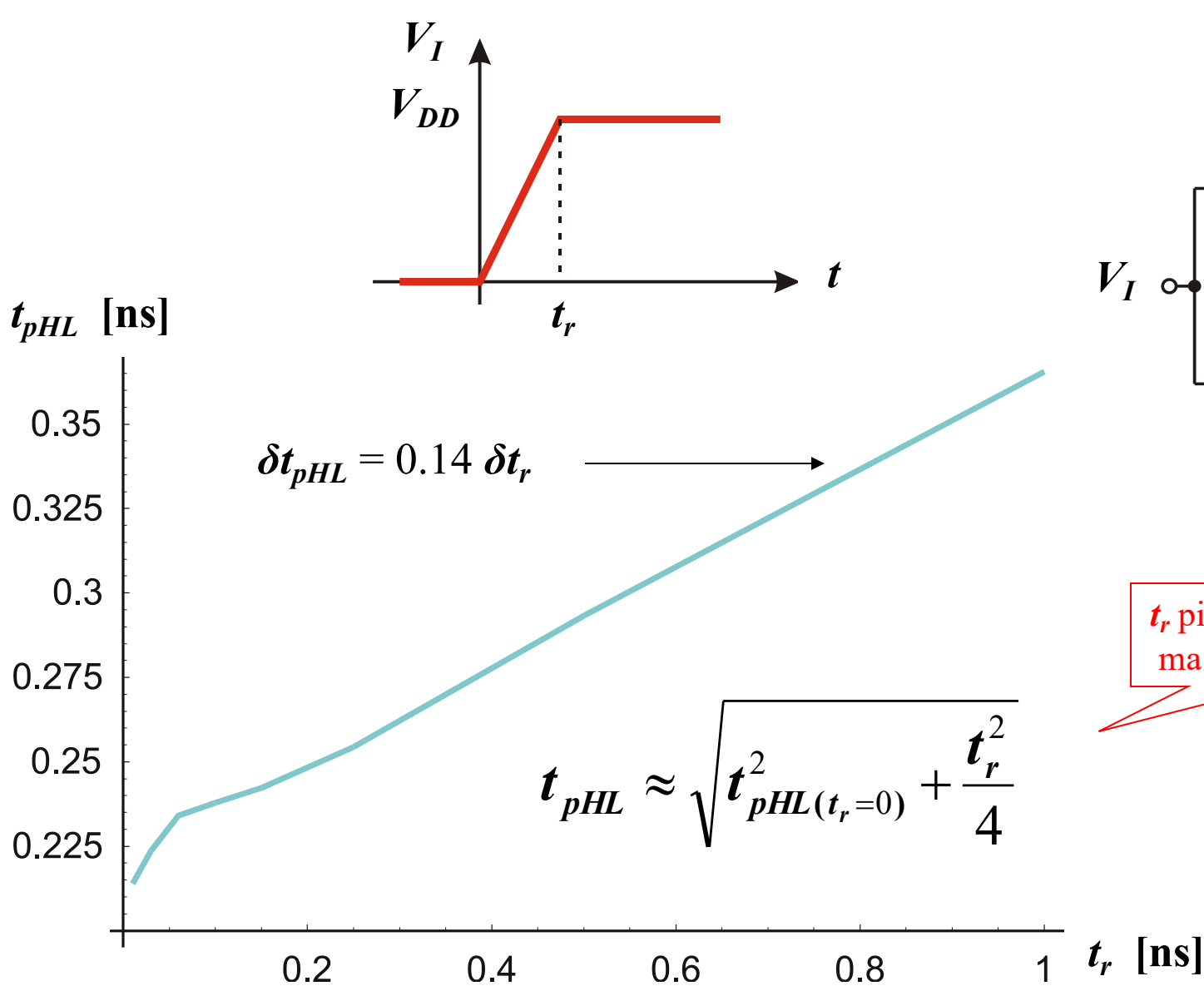
Se $C_W \ll C_{D1} + C_{G3} \rightarrow \alpha_{opt} = \sqrt{\frac{\mu_n}{\mu_p}} = 1.73$

$W_p = 3 W_n \leftarrow NM$ ottimo

$W_p \approx 1.73 W_n \leftarrow t_p$ ottimo

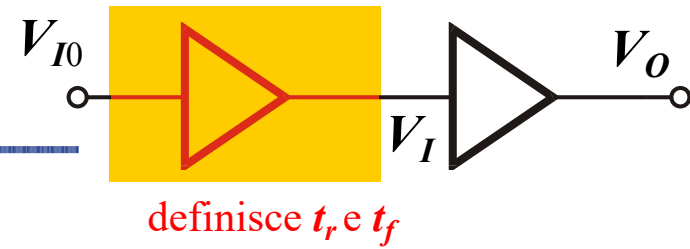
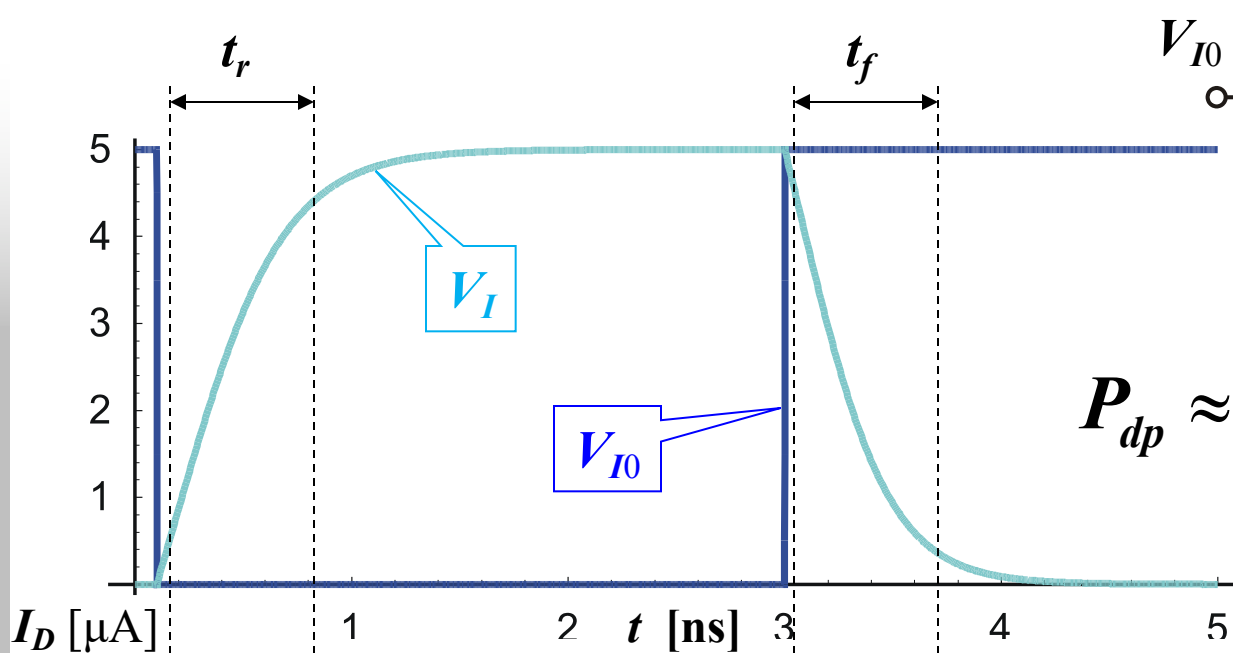




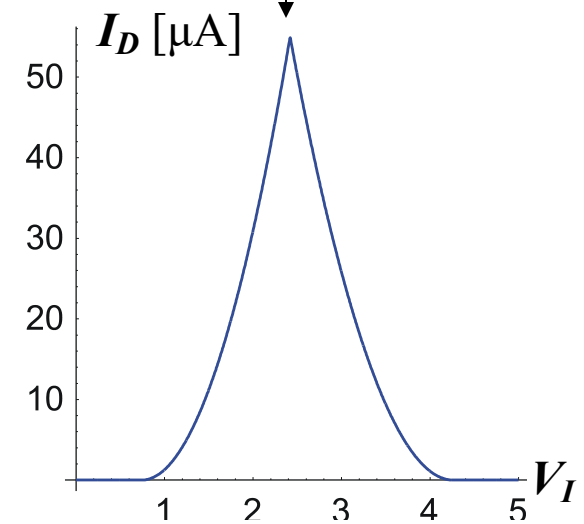
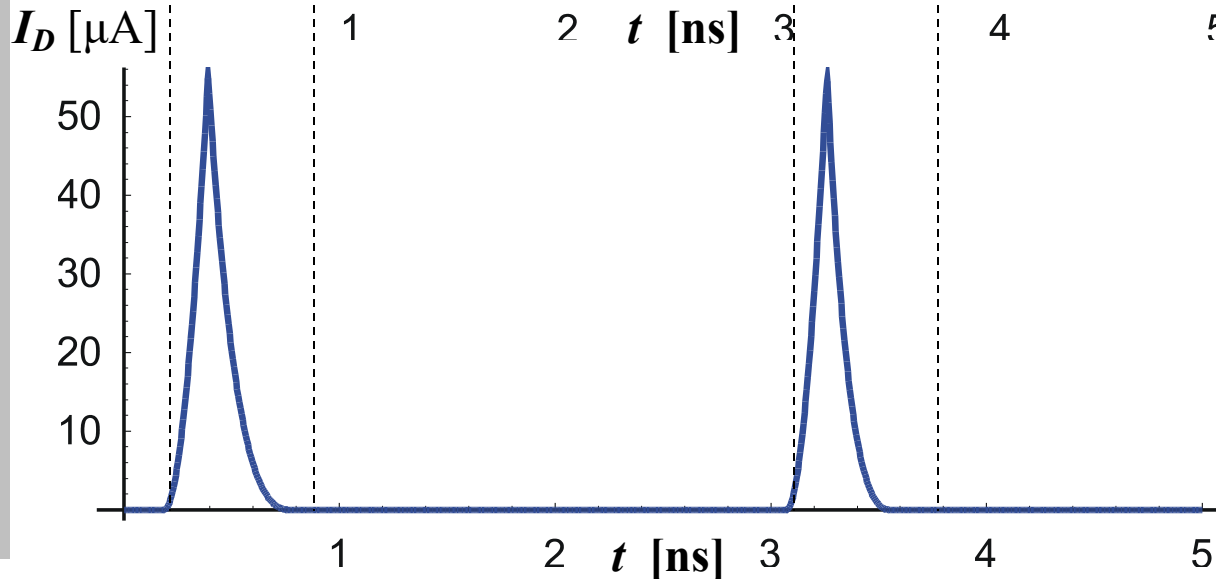


*t_r piccolo aiuta t_p e P ,
ma allarga la banda!*

POTENZA DISSIPATA



$$P_{dp} \approx \frac{t_r + t_f}{2} I_{peak} V_{DD} f$$



POTENZA TOTALE E PDP

normalmente $I_{leakage} < 1\text{nA}$

$$P_{tot} = P_d + P_{dp} + P_s = C V_{DD}^2 f + V_{DD} I_{peak} \frac{t_r + t_f}{2} f + V_{DD} I_{leakage}$$

normalmente domina

$$PDP \approx C V_{DD}^2$$

$$V_{DD} = 5\text{V} \quad C_L = 34.2\text{fF} \rightarrow PDP = 0.86\text{pJ}$$

POTENZA DINAMICA DISSIPATA

$$P_d = C V_{DD}^2 f$$

capacità equivalente

clock

Esempi:

200 000 *gate*, $C = 30\text{fF}/\text{gate}$, 20% attività, $V_{DD} = 5\text{V}$, $f = 100\text{MHz}$

$$P_d = 200\,000 \times 30\text{fF} \times 20\% \times 5^2 \times 100\text{MHz} = 3\text{W}$$

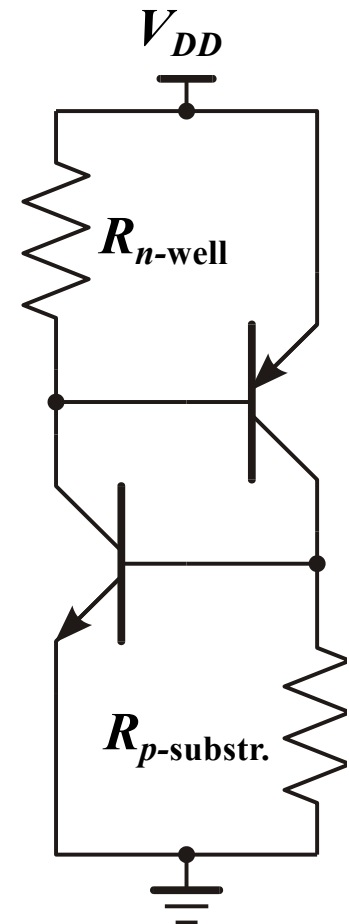
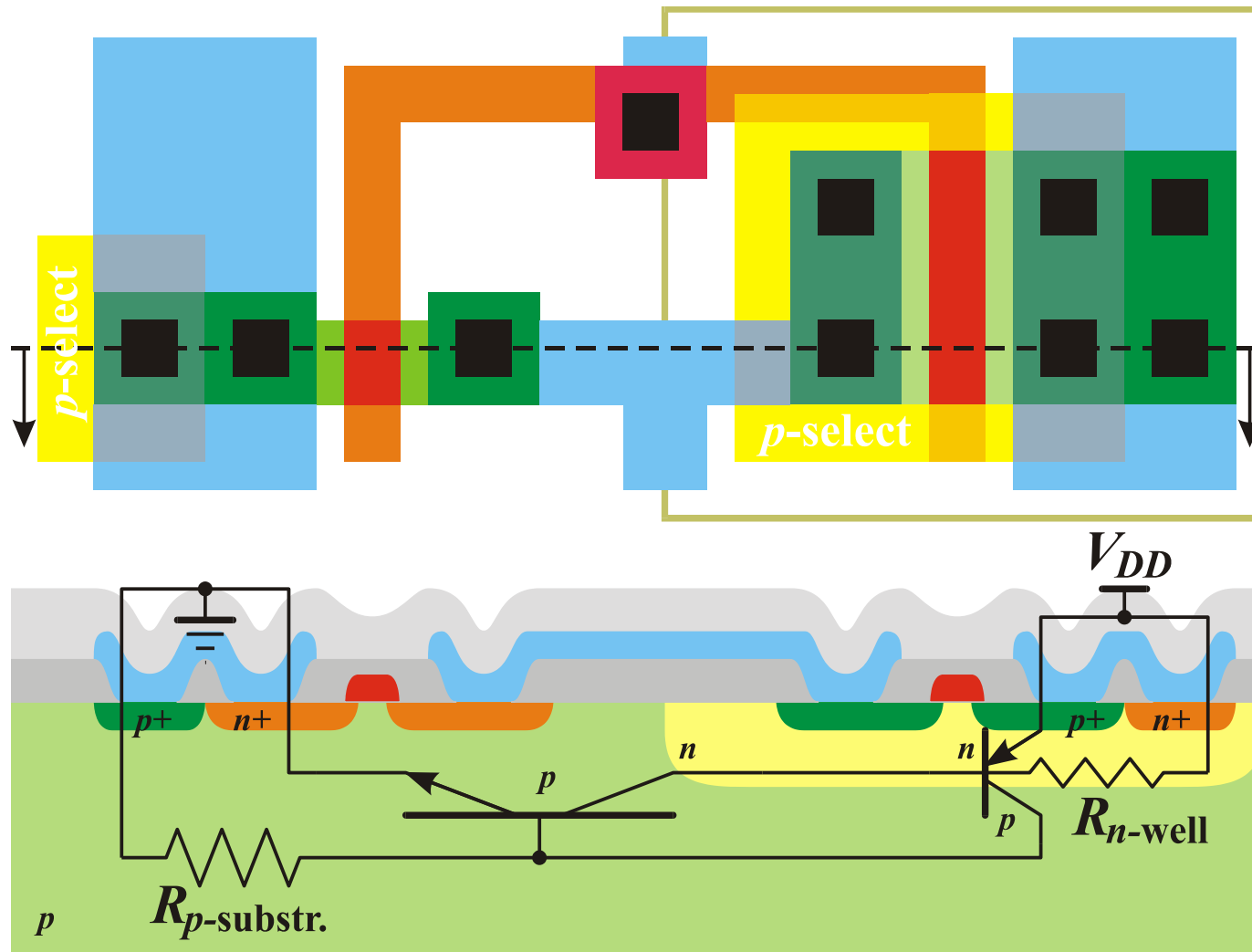
200 000 *gate*, $C = 30\text{fF}/\text{gate}$, 20% attività, $V_{DD} = 3\text{V}$, $f = 100\text{MHz}$

$$P_d = 200\,000 \times 30\text{fF} \times 20\% \times 3^2 \times 100\text{MHz} = 1.1\text{W}$$

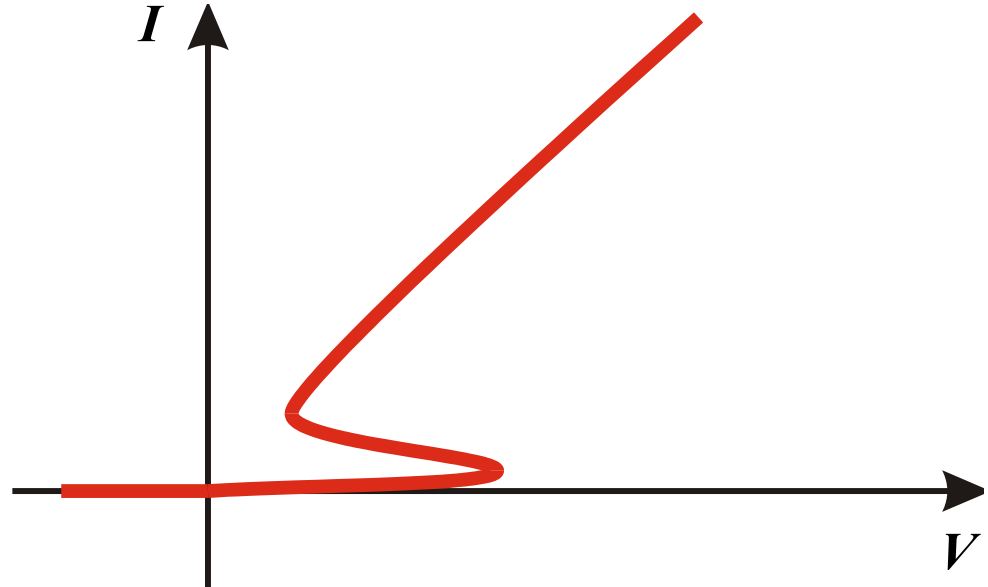
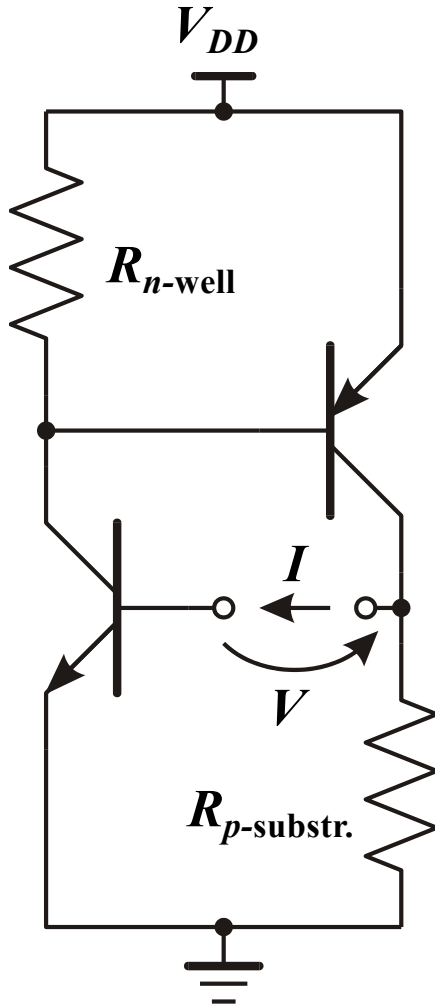
100 *pin*, $C = 20\text{pF}/\text{pin}$, 50% attività, $V_{DD} = 5\text{V}$, $f = 20\text{MHz}$

$$P_d = 100 \times 20\text{pF} \times 50\% \times 5^2 \times 20\text{MHz} = 0.5\text{W}$$

1/3 pin
1/3 rete clock
1/3 calcolo



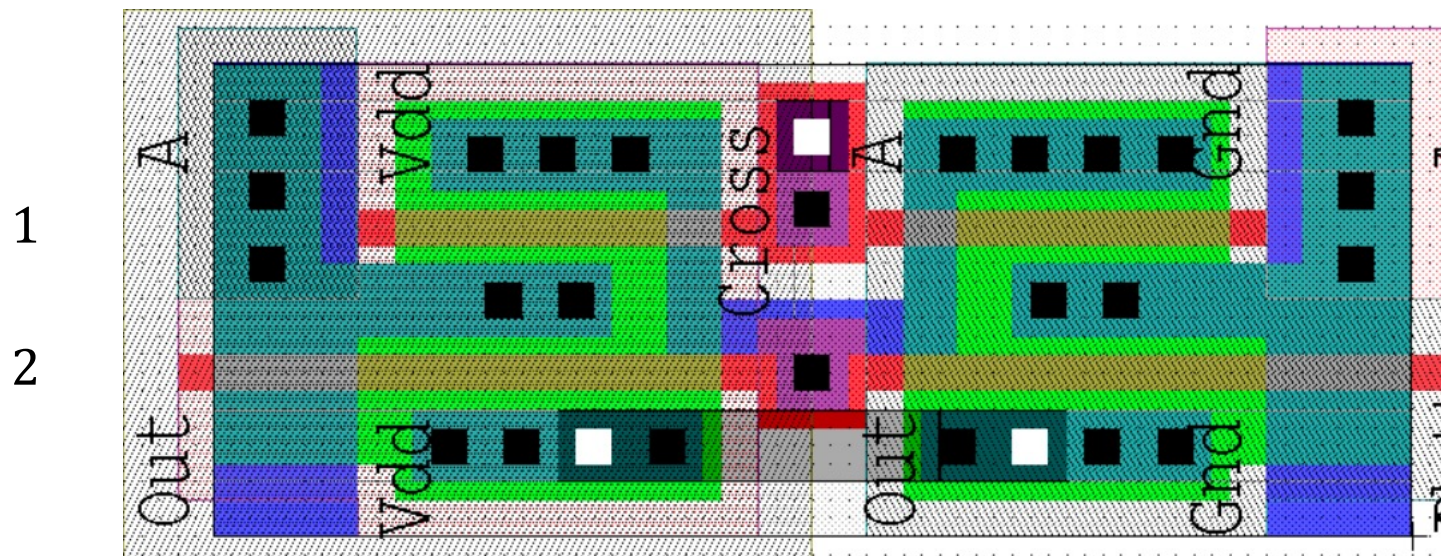
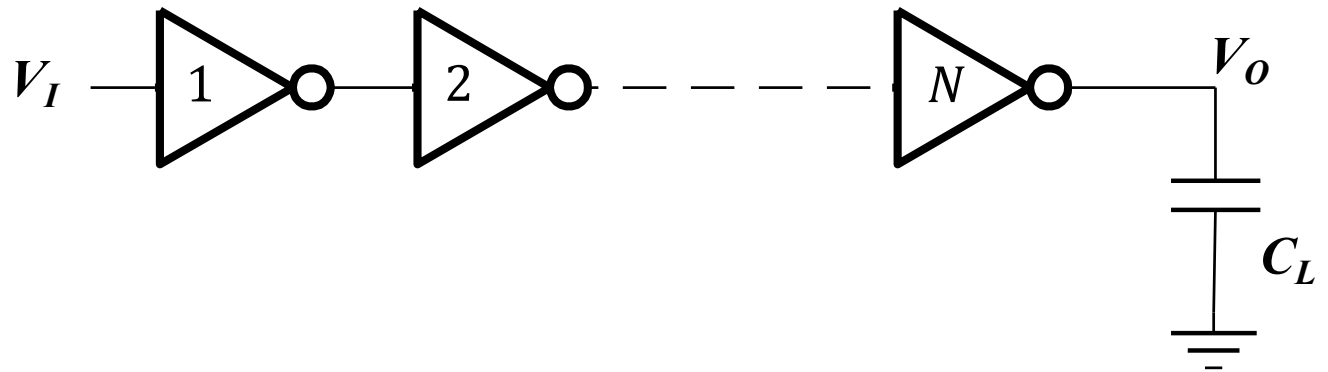
CARATTERISTICA I(V) A S



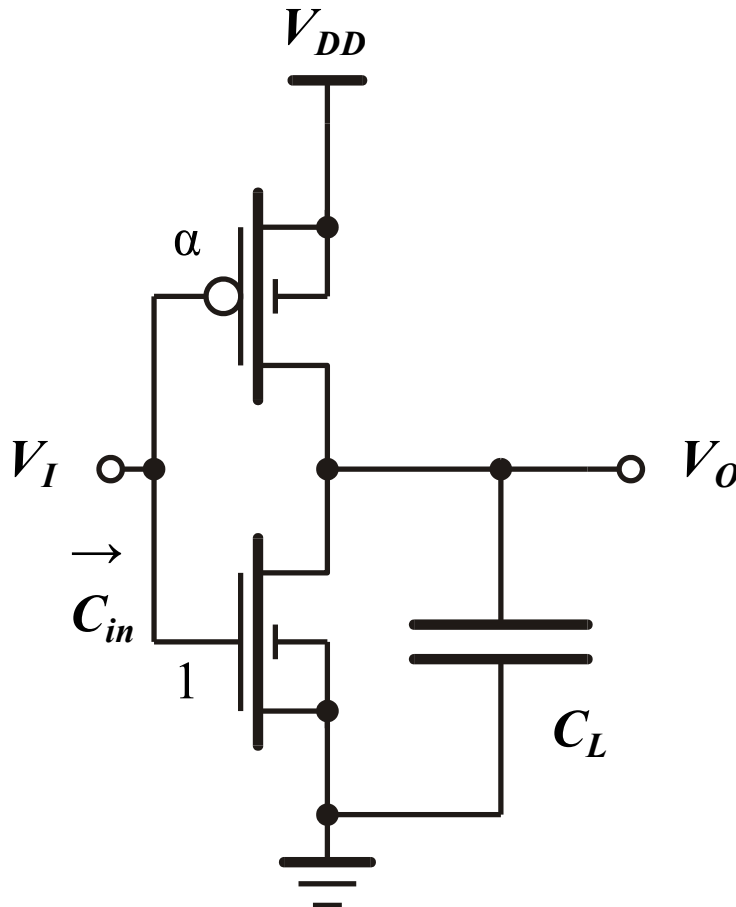
Per ridurre il rischio di *latchup*:

- ✓ Contatto *n-source* e contatto substrato vicini
- ✓ Contatto *p-source* e contatto *n-well* vicini
- ✓ *n-well* di area minima

Dimensionamento Progressivo



Carico Capacitivo Rilevante



$$C_{in} = C_{in\ Min} \approx (1 + \alpha) C_{gate\ Min}$$

Inverter minimo:

$$W_p = \alpha W_{min}$$

$$W_n = W_{min}$$

$$L_p = L_n = L_{min}$$

$$\rightarrow t_p = t_{p0} \text{ se } C_L = C_{in}$$

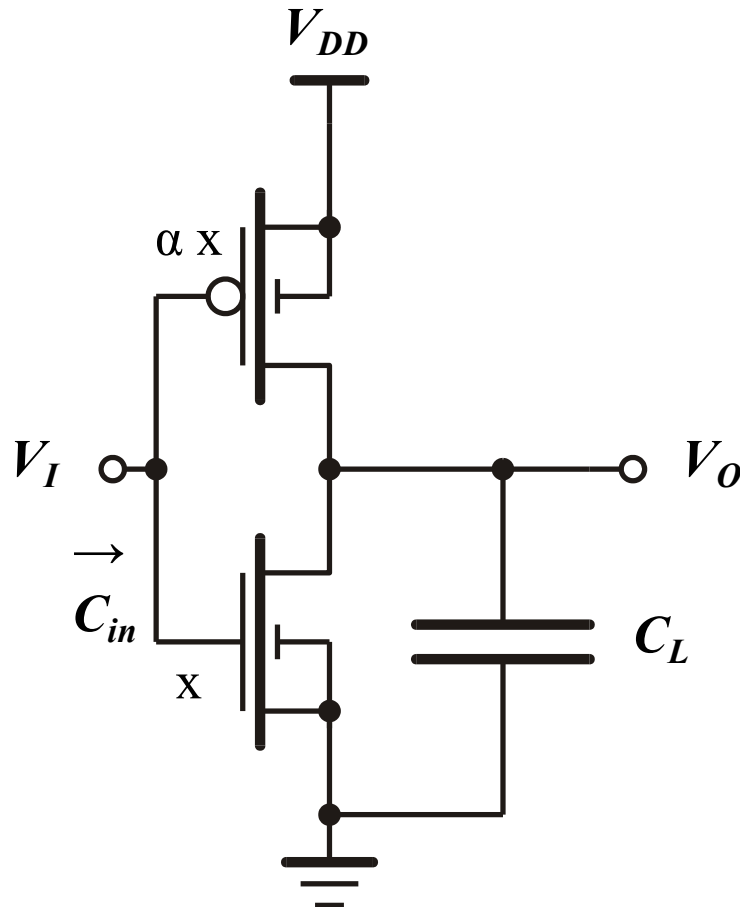
$$C_L = C_{intrinseca\ NOT} + x C_{in}$$

$$\text{se } x \gg 1: C_L \approx x C_{in}$$

$$\rightarrow t_p = x t_{p0}$$

$$\rightarrow t_p = 1000 t_{p0} \text{ se } x = 1000!$$

ad esempio, se il carico è una pista esterna



$$W_p = \alpha \times W_{min}$$

$$W_n = X \times W_{min}$$

$$L_p = L_n = L_{min}$$

$$C_L = X \times C_{in}$$

$$\rightarrow t_p = X \times \frac{t_{p0}}{X} = t_{p0}$$

Come per un inverter minimo

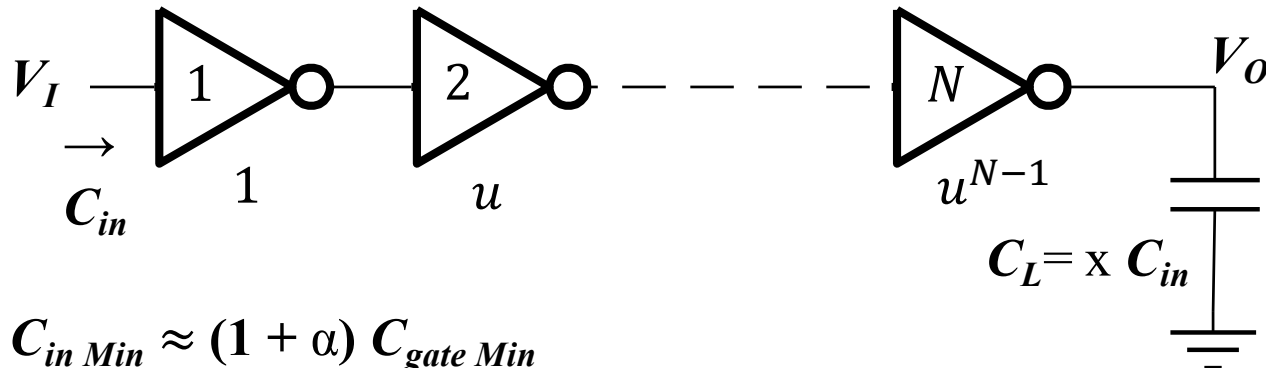
(trascurando la capacità intrinseca)

ma il problema si è spostato a monte!

$$C_{in} \approx X (1 + \alpha) C_{gate\ Min}$$

Dimensionamento Progressivo

Carico a monte minimo:



$$C_{in} = C_{in\ Min} \approx (1 + \alpha) C_{gate\ Min}$$

Hp: $N = 2$ (trascurando la capacità intrinseca: corretto in prog. reale per vincolo area)

$$\rightarrow t_p = t_{p1} + t_{p2} = u t_{p0} + \frac{X}{u} t_{p0}$$

$$\rightarrow \frac{d t_p}{d u} = t_{p0} \left(1 - \frac{X}{u^2} \right) = 0$$

$$\rightarrow u_{ott} = \sqrt{X}$$

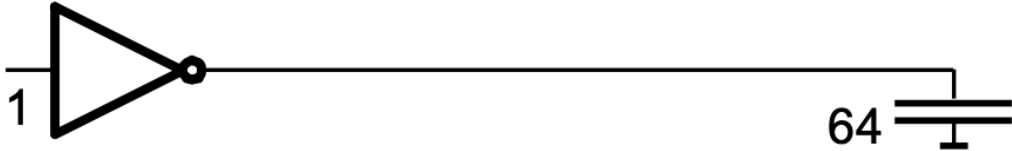
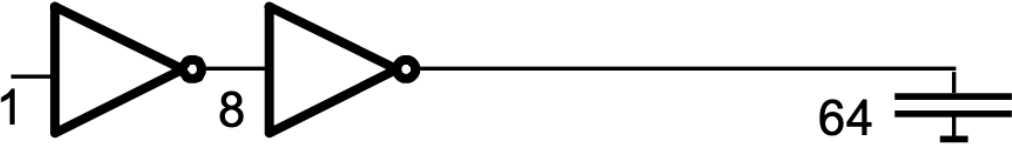
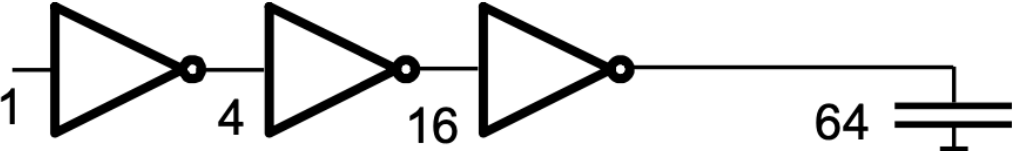
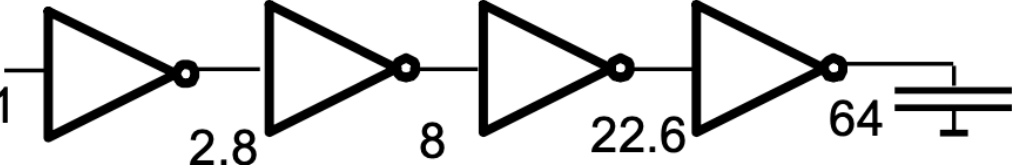
$$\rightarrow t_{p\ ott} = \sqrt{X} t_{p0} + \frac{X}{\sqrt{X}} t_{p0} = 2\sqrt{X} t_{p0}$$

$$\rightarrow u_{ott} = \sqrt[N]{X}$$

$$\rightarrow t_{p\ ott} = N \sqrt[N]{X} t_{p0}$$

Caso ottimo: $u = e = 2.718$, ma porta a N troppo grande e non si usa MAI.

Esempio

	N	$\sqrt[N]{64}$	t_p
	1	64	65
	2	8	18
	3	4	15
	4	2.8	15.3

inoltre, qui la capacità intrinseca non può più essere trascurata
 ($u > \sim 5$ nella pratica, altrimenti si perde troppo in area senza guadagnare con t_p)